

Bachelor in **Applied Data Science**

REAL-TIME SQUAT FEEDBACK USING POSE ESTIMATION AND MACHINE LEARNING

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Abstract

This project presents a tool that provides automated feedback on squat exercises by analyzing pre-recorded videos. Engaging in gym exercises and running can be physically taxing on the human body, and poor posture or incorrect technique often does more harm than good. These issues often arise from a lack of knowledge or adequate guidance. Hiring a trainer is often expensive. In some cases, trainers may also be unavailable.

The objective of this study was to design and develop a system capable of providing real-time feedback on squat performance. The proposed approach utilizes human pose estimation to extract body keypoints from video recordings, combined with deep learning models, including Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, to classify squat types and evaluate posture. In addition, a Large Language Model (LLM) is integrated to generate natural and actionable feedback for users.

The results demonstrate that the system is capable of analyzing squat movements and providing feedback in controlled settings. However, several limitations were identified. The inclusion of the LSTM layer did not improve performance compared to a CNN-only model, suggesting that temporal modeling was not beneficial for this task. Furthermore, the model showed poor generalization to unseen individuals, while performing well on data from participants included in the training set. Posture classification was also found to be more reliable than squat style classification.

Overall, the findings indicate that generalization across individuals is the primary limitation of the system. While effective for same-user feedback, further improvements are required before the system can be reliably applied in real-world scenarios.

Keywords: *LSTM, CNN, MoveNet, Style, Posture*

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1

Introduction

1.1 Introduction

Personal fitness has grown rapidly in recent years, with an increasing number of gyms and fitness facilities becoming widely available. As fitness has become more popular, technology has played a significant role in helping individuals track workouts, monitor progress, and measure performance. Affordable smartphones, cameras, and wearable devices now make it possible to capture real-time data without the need for specialized equipment. However, most of these tools primarily focus on counting repetitions or estimating calories burned. While some AI-powered solutions are effective, there remains a gap between what digital fitness tools can provide and the personalized guidance offered by a professional trainer (Schumann & Doherty, 2024).

Engaging in gym exercises and running can place considerable physical strain on the body, and poor posture or incorrect technique may lead to injury (Bonilla et al., 2022). For example, performing a deadlift with improper form can result in misalignment under heavy load, placing excessive stress on the joints and spine. While many gyms offer personal trainers to support proper technique and reduce injury risk, access to such guidance is often limited. Hiring a trainer can be expensive, and in some cases, trainers may not be available due to location or facility constraints. This highlights the need for accessible systems that can provide real-time feedback to individuals without professional supervision.

Modern smart devices already incorporate basic tracking capabilities. In addition, human pose estimation models such as OpenPose, PoseNet, and MoveNet can efficiently detect

body landmarks, making them suitable for analyzing human movement (Jo & Kim, 2022). Deep learning models, such as Convolutional Neural Networks (CNNs), can be trained to learn both spatial and temporal patterns in movement data, enabling the detection of improper technique and posture.

Beyond motion analysis, Large Language Models (LLMs) offer the ability to generate adaptive, personalized, and context-aware feedback. While many current systems focus primarily on activity recognition and performance statistics, there is significant potential to develop intelligent feedback systems that provide meaningful corrective guidance.

This thesis aims to address this gap by developing a tool that analyzes exercise videos and provides real-time feedback. The proposed system combines human pose estimation with deep learning-based classification to detect improper posture and technique. A language model is further integrated to generate natural, corrective feedback in textual form.

The goal of this project is to improve access to personalized fitness guidance without requiring a human trainer. The scope of the study is limited to a single exercise: the squat.

1.2 Background and Related Work

This section is light review of the relevant research areas to develop a tool for real-time feedback on training. See chapter 2 for more in-depth research.

1.2.1 Squat

The squat is a general term for an exercise that includes multiple variations. In its basic form, it involves bending the knees and lowering the body as if sitting back into a chair, followed by returning to the standing position. Squats are a fundamental training exercise widely used in both rehabilitation and strength training contexts.

The exercise is typically performed using free weights, which are associated with a higher risk of injury. Kerr et al., 2010 reported that 90.4% of weight-training injuries in the United States between 1990 and 2007 were linked to the use of free weights.

A study conducted in Poland investigated students' knowledge of correct squat technique using a questionnaire and a knowledge test. While most participants recognized that improper squat technique and posture can have negative effects on the body, the study found that "over half of the students examined admitted that they do not know what correct squat technique looks like."Czaprowski et al. (2012)

1.2.2 Human Pose Estimation

Human Pose Estimation (HPE) has advanced significantly in recent years with the adoption of deep learning techniques. The goal of HPE is to locate the human body skeleton by

identifying key joints from input data (Zheng et al., 2024). Most modern HPE approaches rely on deep learning, particularly convolutional neural networks (CNNs), to detect body joints from 2D images. By connecting these joints, a skeleton-based representation of the human body can be constructed, which in turn enables the extraction of joint angles, body alignments, and motion trajectories in real time (Dang et al., 2019).

There are several HPE models available. For example, BlazePose is a lightweight convolutional neural network developed by Google (Bazarevsky et al., 2020). Another widely used model is OpenPose, developed by Carnegie Mellon University, which supports multi-person pose estimation (Cao et al., 2021). Additionally, MoveNet is a pose estimation model developed by Google ("Next-Generation Pose Detection with MoveNet and TensorFlow.js", 2021). It provides two variants designed for different use cases, allowing a trade-off between speed and accuracy.

1.2.3 Activity Recognition and Classification

Human Activity Recognition (HAR) focuses on identifying and classifying different human activities. These activities can range from daily living habits and stress levels to structured training exercises and general day-to-day movements (Zhu et al., 2025). Since human behavior is diverse, HAR systems must be capable of recognizing a wide variety of activities.

In the fitness context, activity recognition has been applied to tasks such as exercise counting, intensity tracking, and the classification of common movements like push-ups, squats, and lunges. However, most of these systems prioritize recognition over the quality of execution and corrective guidance. A study by Velloso et al. (2013) demonstrated significant improvement when HAR systems provided feedback on exercise performance, emphasizing proper technique rather than merely identifying the activity.

1.2.4 AI-Driven Fitness Coaching

AI-driven fitness coaching is increasingly widespread, with a variety of applications available to support users during exercise. Many of these applications incorporate tracking systems to monitor progress and performance (Garbett et al., 2021). Some mobile apps provide real-time verbal instructions, corrective pose feedback, and motivational prompts for a fixed set of exercises, while others offer demonstration videos and live-streamed training sessions.

For example, Chen et al. (2023) developed a hybrid CNN–LSTM model to recognize and classify Baduanjin exercise movements. The system utilized OpenPose to extract joint key-points and generate a skeletal representation, which served as input to the model. Similarly, Priyakanth et al. (2025) developed a tool to assist users in performing yoga poses at home. Their system employed MoveNet to predict body landmarks and used TensorFlow for machine learning and data analysis. A webcam captured user movements, enabling the system to provide real-time feedback on posture accuracy.

However, Garbett et al. (2021) also identified several limitations in AI-driven fitness systems. Participants reported cases where the AI failed to recognize correct positions, resulting in inaccurate real-time feedback. Additionally, spatial constraints posed challenges, as these systems often required more physical space for accurate tracking than users typically had available at home.

1.2.5 Feedback

Feedback plays an important role in learning across many domains, including workplace training and physical activity. Effective feedback helps individuals improve performance and refine their skills. It can be delivered in various forms, including audio, text, visual cues, or a combination of these.

A study by Walsh et al. (2009) explored two types of feedback: concurrent feedback (provided during task execution) and terminal feedback (provided after task completion). The study concluded that the effectiveness of each type depends on the nature of the task.

This distinction is directly relevant to exercise contexts such as squat training. Concurrent feedback may assist individuals while they are performing the movement, enabling immediate corrections. In contrast, terminal feedback allows individuals to reflect on their performance after completing the exercise, supporting longer-term learning.

The effectiveness of each approach may also depend on the environment. In high-intensity settings, concurrent feedback may enhance short-term performance, whereas terminal feedback may be more beneficial for long-term skill development.

Large Language Models for Personalized Feedback

Large Language Models (LLMs) have proven to be powerful tools for understanding human expression and generating human-like, context-aware feedback. Their adaptability to individual users makes them highly engaging and effective, leveraging vast amounts of data and large-scale neural networks (Chen et al., 2024).

Some of the most well-known LLMs, such as GPT, LLaMA, and Alpaca, have become instrumental across a wide range of applications, including customer support via chatbots, as well as assistance in healthcare and finance. Despite being trained on vast amounts of data, these models still face challenges in maintaining accuracy and relevance (Ouyang et al., 2023).

1.2.6 Summary

In summary, the squat is a free-weight exercise associated with a higher risk of injury. However, it is generally considered safe when performed correctly with proper technique and

appropriate load. Despite this, many individuals lack knowledge of what constitutes proper squat execution.

Advances in Human Pose Estimation (HPE) and Human Activity Recognition (HAR) enable the recognition and classification of human movements using smart devices. While many existing fitness applications primarily focus on tracking and providing static guidance, recent studies have explored the use of HPE and HAR to guide users through specific exercises. When combined with effective corrective feedback, such systems have the potential to assist individuals in performing squats correctly without the need for a personal trainer. AI-driven systems are increasingly capable of delivering real-time feedback. Furthermore, Large Language Models (LLMs) offer opportunities for personalized, engaging, and motivating coaching, with the potential to enhance both learning and safety during exercise.

By integrating these components, the proposed approach utilizes human pose estimation (HPE) to extract keypoints, which are then processed by a CNN-LSTM model to classify squat types and posture variations. Based on these classifications, predefined feedback templates are selected, and a Large Language Model (LLM) is used to generate coaching feedback. This feedback is delivered in text form to provide real-time guidance to the user

1.3 Problem Statement

Most injuries during weight training are caused by improper technique or poor posture (Velloso et al., 2013). These issues often arise from a lack of knowledge or adequate guidance. Access to in-person coaching from qualified trainers is frequently limited due to factors such as cost, location, or availability.

Most fitness applications that use computer vision or wearable technologies focus on activity tracking, repetitions, and progress monitoring. While some systems provide realtime feedback during training, they still face limitations, such as supporting only a fixed set of exercises or requiring more physical space than users typically have available (Garbett et al., 2021). Although Large Language Models (LLMs) can provide real-time instruction and motivation, challenges such as latency and hallucinations remain, which can compromise the reliability of immediate feedback.

Single-sentence problem statement: This research addresses the challenge of providing users with accurate, real-time feedback on squats through image analysis in a way that is accessible and easy to use.

1.4 Research Objectives

The primary objective of this project is to design and develop a tool that provides real-time feedback to individuals as they engage in physical activity: squat. With the aim of improving

individuals' posture and techniques to reduce injury risk.

In order to achieve this primary objective, the following secondary objectives have been identified:

- Review existing research.
- Implement MoveNet as HPE to extract keypoints.
- Develop an activity classification system to detect squat type and technique errors using classical ML.
- Incorporate a large language model.

1.5 Research Methodology

The methodology in This research utilizes MoveNet to extract 2D keypoints from side-view squat videos. Joint angles and motion-based features derived from these keypoints are used to train a machine learning classifier for detecting squat type and identifying technique errors. When style and posture is detected, template-based feedback is generated and passed to Google Gemini for rephrasing.

First, the dataset will be created and collected by the author. The data will then be preprocessed by downscaling the videos and removing audio. Each recording will subsequently be augmented once using transformations such as scaling, zooming, and shifting, effectively doubling the size of the training dataset from 600 to 1200 samples.

Before applying pose estimation, all videos will be standardized to a fixed length of 50 frames to ensure consistency. MoveNet will then be used to extract the body skeleton and joint keypoints from the input videos. These keypoints will be used to compute features such as knee angle, hip angle, back angle, hip velocity, foot angle, and repetition duration.

Next, a machine learning classifier will determine the squat type (front, back, or goblet) and identify any posture or technique issues, including rounded back, weight on the toes, and insufficient depth, based on the training data. Finally, the detected posture issues and squat style will be matched with predefined feedback templates, which will be rephrased by Google Gemini into clear, context-aware, and motivating instructions for the user.

1.6 Scope and Limits

Scope:

- Real-time analysis of squat exercises from video, including classification of squat types (front, back, and goblet).

- Detection of posture errors with corrective feedback generated using a Large Language Model (LLM).
- Feedback delivered in text form

Out of scope:

- Medical or healthcare advice.
- Professional athlete-level performance analytics
- Multi-person tracking

1.7 Document Structure

This dissertation is organized as follows:

- **Chapter 1: Introduction** Presents the overall motivation, problem statement, research objectives, as well as the scope and limitations of the study.
- **Chapter 2: Literature Review** Reviews existing research on pose estimation, convolutional neural networks (CNNs), their combinations, and the effectiveness of feedback systems.
- **Chapter 3: Methodology** Describes the proposed approach, including the use of MoveNet for keypoint extraction, the applied machine learning models, the system pipeline, and its limitations.
- **Chapter 4: Results/Analysis** This chapter presents the results and evaluates the performance of the proposed system. It includes the experimental results, their analysis, and a discussion of the findings.
- **Chapter 5: Conclusion and Future Work** Summarizes the research, outlines the contributions of the study, evaluates whether the research objectives were achieved, and suggests directions for future work.

2

Literature Review

2.1 Background

Physical activity has been an integral part of human life since the beginning of civilization. Walking, fighting, hunting, and other survival-related actions have always involved physical exertion. Even in ancient Greece, there are records of organized sports and athletic competitions. In modern society, technological advancements such as cars, trains, airplanes, and machines designed to perform manual labor have significantly reduced the need for physical movement in daily life. As a result, obesity and other lifestyle-related health issues have become major public health concerns in the 21st century. Despite this trend, increasing awareness of health and fitness has led to a steady rise in gym memberships and the popularity of gyms as spaces for structured physical activity. Physical activity is a key component of overall health and plays an important role in body-weight regulation, including weight management and weight loss. (Jakicic, 2009). Moreover, regular physical activity improves muscular strength, movement quality, and functional capacity, which are important for reducing injury risk during exercise. Consequently, with the widespread availability of gyms, weightlifting has become one of the most popular forms of physical activity in present-day society.

This paper focuses on the squat exercise, a widely used lower-body movement in both strength training and rehabilitation contexts. Three squat variations are considered: the goblet squat, front squat, and back squat.

2.2 Squat

Squat is a general term for an exercise that has multiple variations. In general, it involves bending the knees and lowering the body as if sitting back into a chair, then standing back up to the starting position. A front squat is performed with a barbell resting on the front of the shoulders (top of the chest), a back squat (free barbell back squat) is done with a barbell placed across the upper back, and a goblet squat is performed while holding a weight (such as a dumbbell or kettlebell) in front of the chest.

Squats are a fundamental training exercise widely used in both rehabilitation and strength training contexts. The movement primarily activates the quadriceps and gluteus maximus, while also engaging the hamstrings, erector spinae, adductor magnus, gastrocnemius, and soleus muscles. Electromyographic studies indicate that muscle activation is greatest during the ascending (concentric) phase of the squat, and that overall lower-limb muscle activation does not differ substantially between front and back squat variations, despite differences in bar position. Importantly, biomechanical analyses have shown that the back squat produces higher knee compressive forces than the front squat, suggesting that the front squat may be a more suitable alternative for individuals with knee-related injuries or joint sensitivity (Clark et al., 2012)(Gullett et al., 2009)(Slater & Hart, 2017).

Squats are generally considered a safe and effective exercise when performed with proper technique and appropriate load. The squat is typically performed using free weights, and studies show that indicates that free-weight exercises account for the majority of weight-training-related injuries. Kerr et al., 2010 reported that 90.4% of weight-training injuries in the United States between 1990 and 2007 were associated with the use of free weights. These injuries most commonly resulted from dropped weights, overexertion, or loss of balance. Together, these findings suggest that injuries associated with squat training are often preventable through proper technique instruction, progressive load management, and adequate supervision, highlighting the importance of feedback and monitoring during training.

This is one of the main reasons personal trainers are available at gyms, to educate and guide individuals during their workouts. However, personal trainers can be expensive, not always affordable, or may not be available geographically. A study conducted in Poland investigated students' knowledge of correct squat technique using a questionnaire and a knowledge test administered to 412 physical education students. The results showed that nearly 71% of the students recognized that incorrect squat technique and posture may have negative consequences for the on the body. However, despite this awareness, the authors reported a substantial lack of technical knowledge, as stated in the study Czaprowski et al., 2012: "Over half of the students examined in the test admitted that they do not know what the correct squat looks like."

An analysis of training data from millions of workouts logged in the StrengthLog app showed that the squat was the second most popular strength training exercise overall, with the bench

press ranking first. ("The Most Popular Gym Exercises for Men & Women in 2025 – Strength-Log", n.d.).

The correctly performed squat from Czaprowski et al., 2012:

- head – in a neutral position (gaze directed horizontally or slightly up)
- thoracic segment of the spine – slightly straight, mobile
- lumbar segment of the spine – in a neutral position, stable
- hip joints – bent, moved backwards, mobile
- knee joints – set in a line between hips and feet, stable
- feet – set at the width of shoulders, in a neutral position or 7°-10° aside, heels touching the floor, stable
- the trunk angle during the squat should be stable in every phase
- optimal depth of the squat: the angle of the bend in a knee joint – 90°- maximal depth of the squat: 115°-125° (thighs are parallel to the floor)
- the performance of a squat is commenced from a bend in hip joints and then in knee joints.

2.3 Models

2.3.1 Pose estimation

In recent years, human pose estimation has gained significant attention across various fields, including sports analytics, gaming, and realistic animation, where accurate capture of human movement is essential. The goal of pose estimation models is to locate human body parts and construct a skeletal representation by identifying key joints, commonly referred to as keypoints or landmarks, from input data such as images or videos. (Zheng et al., 2024).

One such pose estimation model is BlazePose, a lightweight convolutional neural network that predicts 33 key landmarks on the human body and utilizes a pose tracking network. It is designed to be fast and effective on mobile devices, supporting temporal tracking and operating at approximately 30–60 fps on mobile hardware. Its architecture consists of a lightweight body pose detector that first identifies the person's body, followed by a pose tracking network that estimates the 33 body keypoints across frames (Bazarevsky et al., 2020). BlazePose is the human pose estimation model integrated into MediaPipe, a multimedia processing framework developed by Google (Lugaresi et al., 2019).

Another well-known pose estimation framework is OpenPose, developed by researchers at Carnegie Mellon University (CMU). OpenPose is a real-time, multi-person pose estimation system capable of predicting up to 135 keypoints (landmarks) on the human body. It employs a bottom-up architecture, in which confidence maps are first used to detect individual joint locations across all people in an image. These joints are then associated using Part Affinity Fields (PAFs), which are two-dimensional vector fields that encode the spatial relationships between body parts. Based on these associations, the detected joints are assembled into full human skeletons, and individual persons are automatically identified and separated (Cao et al., 2021).

MoveNet is another pose estimation model developed by Google, designed with a strong focus on both speed and accuracy. It predicts 17 key body landmarks and is available in two variants: one optimized for mobile and desktop devices, and another designed for latency-critical applications requiring real-time performance. MoveNet estimates 2D keypoints and can support multi-person pose estimation, depending on the specific model configuration. However, it is a frame-based approach and does not perform explicit temporal tracking of the human body across frames. (“Next-Generation Pose Detection with MoveNet and TensorFlow.js”, 2021).

2.3.2 CNNs

3D Convolutional Neural Networks (3D CNNs) are generally preferred for video classification tasks. Unlike 2D CNNs, which operate over the spatial dimensions of height and width and are better suited for image analysis, 3D CNNs extend convolution operations into the temporal dimension as well. This allows them to learn how objects and people change and move over time, capturing both spatial and temporal features within video data. Another notable architecture is the R(2+1)D model, developed by Tran et al. (2017), which decomposes the traditional 3D convolution into separate 2D spatial and 1D temporal convolutions. This design allows the model to first learn what an object looks like through spatial feature extraction and then learn how it moves over time through temporal modelling. This separation not only improves the model’s ability to capture motion dynamics but also reduces the computational complexity compared to standard 3D CNNs, making it more efficient for action recognition tasks in videos. Another commonly used model is the Long Short-Term Memory (LSTM) network. While CNNs are effective at capturing spatial features, LSTMs are designed to understand how data changes over time. Because of this complementary relationship, CNNs and LSTMs are often combined for video-based tasks.

For example, Chen et al. (2023) developed a hybrid CNN–LSTM model to recognize and classify Baduanjin exercise movements from video data. The architecture uses a pre-trained VGG16 to extract visual features from each video frame and a long short-term memory network (LSTM) to learn how those features change over time for action classification. For comparison, the authors also implemented a skeleton-based baseline method that employs

OpenPose to extract joint keypoints and manually engineered geometric features, which are subsequently classified using an LSTM network.

2.3.3 Combinations

Pose estimation combined with deep learning techniques, has been increasingly applied in sports and physical activity analysis to classify movements, evaluate performance, and predict performance-related outcomes (Moustakas, 2025). These methods enable objective, video-based analysis of human movement and support scalable applications in real-world training environments.

A project conducted by students at the Noroff School of Technology and Digital Media utilized MoveNet Thunder as a pose estimation model to extract keypoints from side-view video recordings of deadlifts. This was followed by either a keypoint-based LSTM model or a video-based R(2+1)D CNN, after which a two-stage classification approach was applied. In this setup, the first stage classified the type of deadlift, and the second stage evaluated the execution form. Based on the classification results, predefined templates were sent to Gemini to provide users with corrective feedback. The study compared the long short-term memory network (LSTM) model, which captures movement patterns over time while being less affected by clothing, lighting, and background, with the R(2+1)D CNN, whose core design focuses on applying 2D spatial convolutions per frame followed by 1D temporal convolutions. The results showed that the LSTM model achieved the best performance in style classification, while the R(2+1)D CNN performed better in execution classification. Additionally, the students evaluated their LSTM model against a 3D CNN trained both with and without keypoint annotations. Across these comparisons, the LSTM model achieved a higher average F1-score, demonstrating the effectiveness of keypoint-based temporal modeling for this task (Grymyr & Lawal, 2025).

Another project conducted by Guo et al. (2021) utilized dance videos from students participating in cheerleading courses to develop a tool for analyzing physical coordination. The objective of the study was to design a video-based pose analysis and visualization system capable of automatically evaluating physical coordination in cheerleading and dance training contexts. To achieve this, the authors proposed a custom pose estimation architecture named EasyNet and EasyNet+. The approach follows a top-down, single-person, 2D pose estimation approach based on heatmap-based keypoint detection, using a ResNet backbone as the CNN feature extractor. EasyNet performs the core neural-network-based pose estimation, while EasyNet+ extends this framework with a rule-based temporal and anatomical correction module to improve robustness and temporal consistency in video sequences. The study evaluated pose estimation quality in dance videos using the Percentage of Correct Frames (mPCF) metric. Results showed that EasyNet+ achieved an 85.97% correct-frame rate on challenging real-world cheerleading and dance videos.

Similarly, Priyakanth et al. (2025) developed a browser-based tool to assist users in perform-

ing yoga poses at home. The application uses a webcam to capture the user's live video stream and overlays a skeletal visualization for pose guidance. The system architecture employs MoveNet, based on a MobileNetV2 backbone with a Feature Pyramid Network (FPN), for real-time 2D human pose estimation, producing heatmap-based predictions of the person center and body keypoints. For pose recognition, a k-Nearest Neighbors (k-NN) classifier is used to compare the live pose feature vector with stored pose examples and output a pose match confidence. A rule-based decision mechanism is applied, where a similarity exceeding 95–97% results in a green skeleton and the activation of a pose timer. Otherwise, the skeleton remains white. The authors report that the Yoga Mentor system can reliably detect and guide seven yoga poses in real time, providing immediate visual feedback to support correct pose execution.

2.4 Feedback

When performing weight training exercises such as squats, maintaining correct posture and technique is crucial to prevent injury. Therefore, incorporating a feedback system into a physical activity classification model can be highly effective in assisting users and promoting safer exercise execution.

Feedback can be delivered in various forms, including audio, textual, visual, or a combination of these modalities. It can also be provided during the task (concurrent feedback) or after task completion (terminal feedback). Walsh et al. (2009) study showed that terminal feedback was more effective in promoting learning, as evidenced by superior performance on transfer tests compared to concurrent feedback. The authors further concluded that the relative effectiveness of concurrent and terminal feedback depends on the nature of the task, with concurrent feedback potentially enhancing short-term performance while impairing long-term learning and skill transfer. In dynamic or high-intensity environments, learners often have limited cognitive capacity to process and reflect on feedback in real time, which may further reduce the effectiveness of concurrent feedback. While verbal feedback delivered during exercise execution can support immediate error correction, delayed or visual feedback (such as reviewing one's performance using mirrors or video recordings) may facilitate deeper reflection and consolidation of movement patterns. This reflective process can ultimately support more robust long-term technique improvement.

Another study examining the effectiveness of feedback was conducted by Velloso et al. (2013), who employed activity recognition techniques to enhance the quality of execution in weight-lifting exercises by providing real-time feedback during training. The authors aimed to improve execution quality in two specific exercises by investigating two complementary architectures, a sensor-based machine-learning approach using wearable inertial sensors combined with a Random Forest classifier, and a model-based approach that represented exercises through a layered, rule-based architecture operating on joint positions obtained from a Kinect sensor. In the study, participants performed the exercises both with and without

the feedback system enabled. The results demonstrated significant improvements in execution quality when feedback was provided, with biceps curls in particular showing a reduction of approximately 79% fewer execution mistakes.

2.5 Summary

Research on weight training continues to evolve, with technology playing an increasingly important role in enhancing physical activity analysis and performance assessment. Proper guidance in the execution of training exercises is, and will remain, a critical factor in the prevention of injuries and in supporting both novice and experienced individuals.

In the case of the squat, a free-weight exercise, the risk of injury is elevated when performed without proper technique. The movement engages multiple muscle groups simultaneously while requiring the body to stabilize an external load. The squat is one of the most popular exercises, it is generally considered safe and effective only when executed with correct technique and an appropriate load.

Human pose estimation has gained significant attention across various fields, including sports analytics, gaming, and realistic animation, due to its ability to accurately capture human movement. As a result, it serves as a valuable tool for analyzing how individuals perform exercises. By predicting key landmarks on the human body, pose estimation models can track movement over time. A wide range of pose estimation methods exists, each designed for different applications, such as multi-person tracking, full-body motion capture, or facial landmark detection, depending on the user's needs.

Combining pose estimation with learning models such as convolutional neural networks (CNNs), which can be trained on specific movement patterns, represents a promising approach for guiding individuals in their training. Three-dimensional CNNs (3D CNNs) are capable of learning both spatial features and temporal motion patterns. Additionally, models such as long short-term memory networks (LSTMs) are specifically designed to capture how data evolves over time. Integrating these models with pose estimation enables the system to learn representations of correct exercise execution, making it possible to assess performance quality.

Furthermore, integrating a feedback system that delivers guidance through various ways such as audio, text, visual cues, or a combination of these, can help users correct mistakes and reduce the risk of injury.

Building on these studies, particularly Grymyr and Lawal (2025), this work is inspired by the development of a tool designed to classify squat exercises and evaluate execution technique. The proposed system utilizes human pose estimation to capture key body landmarks and employs convolutional neural networks (CNNs) and long short-term memory networks (LSTMs) to learn and recognize both squat types and execution quality. Additionally, a large

language model (LLM) is integrated to generate coaching feedback, enabling the system to provide users with real-time, actionable guidance during training.

3

Methodology

This project presents a tool that provides automated feedback on squat exercises by analyzing pre-recorded videos. The system utilizes pose estimation via MoveNet Thunder, classifies the type of squat, detects posture issues, and generates feedback using predefined templates through the Gemini framework. This section outlines the research design, data collection and preprocessing, the models used, and also discusses the limitations.

3.1 Research design

This study employs a quantitative research design, focusing on the collection and analysis of structured video data of squat exercises. The primary objective is to develop and evaluate a system capable of accurately classifying squat types and identifying common posture errors. The design incorporates elements of experimental research, as participants' squat movements are systematically recorded under controlled conditions. The resulting videos are then processed through a computational pipeline to extract quantitative keypoints for further analysis and model training.

The research design includes the following key components:

1. Data Acquisition: Collection of a structured dataset of squat videos.
2. Data Preprocessing: Standardization of video resolution, frame rate, and sequence length, followed by pose estimation using MoveNet Thunder.

3. Data Analysis: Classification of squat type and posture evaluation using 1D Convolutional Neural Networks (1D-CNNs) combined with Long Short-Term Memory (LSTM) layers, followed by automated feedback generation through Gemini templates.
4. Gemini Rephrasing: The selected template is sent to Gemini, which generates a natural-language message while preserving the core corrective instructions.
5. Validation and Reliability: Procedures to ensure the consistency and accuracy of both data collection and model predictions.

3.2 Data collection

3.2.1 Data recording

The squat dataset will be developed by the author. It will be created by recording side-view videos using a Samsung Galaxy Z Flip 6. A side-view perspective is chosen to ensure that joint visibility remains consistent across all samples and to provide a reliable and informative representation of movement patterns and posture deviations. The phone will be mounted on a tripod placed on a table to ensure a stable and straight view of the exercises. Each recording will be saved in MP4 format and will contain 50 repetitions for each of the three squat types: front, back, and goblet squats and the four posture variations: correct, rounded back, insufficient depth, and weight on toes.

All videos will be filmed in the same room, using masking tape to mark the positions of the table, tripod, and the author to ensure consistent placement across all recordings. Filming in a controlled indoor environment will provide stable lighting and a plain background, reducing detection noise and ensuring uniform recording conditions. The author will wear the same tight-fitting clothing throughout the sessions to maintain clear visibility of body movements. The author is a male in his 30s, and it is acknowledged that this may introduce certain limitations or biases in the dataset.

3.2.2 Data preprocessing

The recordings will be captured at 30 frames per second (fps) with a video resolution of 1440 × 1504 pixels. Each recording will then be split into 50 individual video clips, each lasting between 1 and 3 seconds, and downscaled to a resolution of 540 × 540 pixels. Microsoft Clipchamp is chosen for this step due to its ease of use and the author's familiarity with the software. This process is expected to reduce the file sizes to a range of approximately 200 to 1200 KB. Subsequently, all clips will be resized to 480 × 270 pixels, and the audio tracks will be removed using FFMPEG. The audio will be removed both to reduce storage requirements and to eliminate unnecessary data that does not contribute to pose estimation or classification.

The downscaling will also be performed as a precautionary measure, as previous studies have reported delay and performance issues when training models on higher-resolution videos (Grymyr & Lawal, 2025). After all processing steps are completed, the final file sizes are expected to be reduced further to between 25 and 75 KB per video, making the dataset more storage-efficient and easier to load during training.

3.2.3 Data augmentation

After all the videos have been recorded, clipped, and resized, the dataset will consist of 600 videos. Each recording will then be augmented using rotation, scaling, and shifting techniques to increase dataset variability and improve model generalization. Specifically, each video will be randomly rotated between -10° and $+10^\circ$, zoomed in or out by 4–6%, and shifted horizontally and vertically by $\pm 5\%$ of the frame's width and height. After augmentation, the resulting recordings are expected to range in size from approximately 75 to 150 KB.

This augmentation will be performed to increase the size of the dataset, each recording will be augmented once, resulting in a total of 1,200 videos, effectively doubling the dataset. Providing each squat type and posture variation with a balanced set of 100 videos ensures comparable representation across categories and helps reduce bias during training.

The augmentation will be performed with carefully chosen scales to ensure that all joints remain visible for MoveNet during rotation. Zooming will be applied cautiously to avoid cropping any limbs, while translations will be used to simulate realistic variations in user alignment.

3.2.4 Dataset Structure and Labels

All recordings will be placed in a hierarchical folder structure reflecting posture deviation and squat type and labeled according to their type and posture variation using the following format: "Front_corr", "Front_id", "Front_rb", and "Front_toe". Table 3.1 provides a summary of the dataset.

3.3 Pose Estimation

Pose estimation is a crucial step in this study, as it transforms raw video frames into structured numerical representations of the human body. For this purpose, MoveNet Thunder was selected as the pose estimation model.

3.3.1 Why MoveNet Thunder

MoveNet is a highly accurate pose estimation model capable of detecting 2D joint locations across 17 keypoints, including major body joints such as the shoulders, elbows, hips,

Table 3.1: The class distribution of the generated dataset.

File Name	Style	Number of recordings
Back_corr_n	Back Squat Correct	100
Back_id_n	Back Squat Insufficient Depth	100
Back_rb_n	Back Squat Rounded Back	100
Back_toe_n	Back Squat Weight On Toes	100
Front_corr_n	Front Squat Correct	100
Front_id_n	Front Squat Insufficient Depth	100
Front_rb_n	Front Squat Rounded Back	100
Front_toe_n	Front Squat Weight On Toes	100
Goblet_corr_n	Goblet Squat Correct	100
Goblet_id_n	Goblet Squat Insufficient Depth	100
Goblet_rb_n	Goblet Squat Rounded Back	100
Goblet_toe_n	Goblet Squat Weight On Toes	100
Total =		1200

knees, and ankles. This high level of accuracy is essential for identifying small posture deviations in short video sequences. The Thunder variant was chosen over Lightning because, although Lightning is faster, Thunder provides greater precision. In the context of this controlled dataset, accuracy was considered more important than real-time inference speed for this study (“Next-Generation Pose Detection with MoveNet and TensorFlow.js”, 2021).

MoveNet Thunder is designed for single-person pose estimation, which aligns well with the dataset, as each video will only contain one person. It is also robust to variations in resolution, making it effective even when videos are downscaled, since it can maintain accurate joint detection despite having fewer pixels.

3.3.2 Preprocessing for Pose Estimation

Before processing any of the videos with MoveNet Thunder, it is necessary to ensure consistency and optimal keypoint detection. The first step will involve using OpenCV to extract all frames from the videos, followed by resizing and padding to prepare the data for pose estimation. Each frame will be resized to 256×256 pixels with appropriate padding to preserve the original aspect ratio, as MoveNet was trained on square input dimensions and any distortion could negatively affect keypoint accuracy. Additionally, the color channels will be converted from BGR (OpenCV’s default format) to RGB, which is the format required for MoveNet to function correctly.

3.3.3 Keypoint Detection

MoveNet Thunder detects 17 keypoints in each frame: Nose, Left Eye, Right Eye, Left Ear, Right Ear, Left Shoulder, Right Shoulder, Left Elbow, Right Elbow, Left Wrist, Right Wrist,

Left Hip, Right Hip, Left Knee, Right Knee, Left Ankle, and Right Ankle. Each keypoint is represented by a (y, x, confidence) array, resulting in a (17, 2) output for every frame. For the temporal CNN-LSTM, only the x and y coordinates will be used as input features.

3.3.4 Variable Frame Lengths

The dataset will consist of video recordings ranging from 1 to 3 seconds (approximately 30–90 frames). Since temporal models require fixed-length sequences, all recordings need to be standardized to a fixed length of 50 frames. This value was chosen because it lies roughly in the middle of the expected frame range: all recordings are expected to contain at least 30 frames, while none will exceed 90 frames. Videos with fewer than 50 frames will be linearly interpolated, and videos exceeding 50 frames will be downsampled to meet the fixed-length requirement. This process ensures that all input tensors maintain a consistent shape of (50 frames $\times$ 17 keypoints $\times$ 2 coordinates) for model training.

3.4 Model Architecture

The neural network developed for this study will be designed to classify squat types and detect posture deviations. It will take as input the temporal sequences of keypoints extracted by MoveNet Thunder. After preprocessing, each video will be represented as a tensor of shape (50, 34), corresponding to 50 time steps with 34 spatial features (derived from 17 keypoints $\times$ 2 coordinates).

The model follows a hybrid temporal architecture, combining a 1D Convolutional Neural Network (1D-CNN) to capture short-term local motion patterns with a Long Short-Term Memory (LSTM) network to model longer-term temporal dependencies across the full squat sequence. This structure enables the model to learn both pose transitions and movement dynamics, which are essential for differentiating between correct and incorrect squat techniques.

3.4.1 Layer Choices

The model will consist of six layers, including a temporal convolution layer, a pooling layer, a recurrent layer, a shared dense layer, and two task-specific output layers. See table See table 3.2

The input layer will receive only the temporal sequences of keypoints. The 1D convolutional layer will use a kernel size of 3, allowing the model to capture short-term motion patterns across consecutive frames. A total of 32 filters has been selected to provide sufficient feature extraction capacity while remaining lightweight, an appropriate balance given the relatively small dataset. A max-pooling layer will be included to reduce the risk of overfitting by downsampling the feature maps and emphasizing the most informative features.

An LSTM layer is then used to model long-range temporal dependencies, which are essential for meaningful posture assessment across the entire squat motion. The LSTM comprises 64 units, chosen to provide adequate temporal memory without excessively increasing model complexity.

A shared dense layer follows, reducing the extracted features into a compact representation and helping to discard noise. Finally, two separate softmax output layers, one for squat type and one for posture classification, enable multitask learning, allowing the network to predict both labels simultaneously.

Table 3.2: Model architecture used for squat type and posture classification.

Layer	Type	Units / Filters	Input	Output
Input	-	(50, 34)	Raw keypoint sequence	(50, 34)
Conv1D	1D-CNN	32 filters, kernel size = 3	(50, 34)	(48, 32)
MaxPool1	1D max pooling	Pool size = 2	(48, 32)	(24, 32)
LSTM	LSTM	64 units	(24, 32)	(64,)
Dense_shared	Dense	32 units	(64,)	(32,)
Output_squat	Dense	3 units, softmax	(32,)	(3,)
Output_posture	Dense	4 units, softmax	(32,)	(4,)

3.5 Real-Time Feedback

After the classification model is trained, the tool require a real-time feedback component. Gemini has been selected as the natural-language feedback system, using structured templates to generate clear and consistent coaching instructions.

3.5.1 Why Gemini

Gemini was selected to enhance the naturalness and fluency of the feedback. Traditional feedback systems often lack human-like phrasing, but Gemini can rephrase predefined templates to generate more coherent and readable sentences for the user. While the system is constrained to these predefined templates, this approach ensures that all feedback remains consistent, safe, and biomechanically accurate, while still allowing for natural-sounding rewording. Using Gemini eliminates the need to manually write hundreds of natural-language variations for all squat types and posture faults. Furthermore, this approach supports scalable adaptation, allowing new squat types or posture issues to be added without extensive manual rewriting of feedback content.

3.5.2 Pipeline

Gemini is the final model used in the workflow, bridging pose estimation and user feedback. The complete pipeline is as follows:

1. Pose Estimation: MoveNet Thunder generates 17 keypoints for each frame, representing the user's body joints.
2. Temporal Analysis: The hybrid 1D-CNN + LSTM model analyzes the sequences of keypoints to determine the squat type and posture status
3. Template Matching: Based on the identified squat type and posture status, the system selects the appropriate predefined feedback template.
4. Gemini Rephrasing: The selected template is sent to Gemini, which generates a natural-language message while preserving the core corrective instructions.
5. Feedback Delivery: The rephrased feedback is returned to the user interface for real-time guidance.

3.6 Feature Representation

Each video sequence will be represented as a 2D array of shape (50, 34), corresponding to 50 time steps and 34 features (derived from 17 joints $\times$ 2 coordinates). No additional handcrafted features will be used. Instead, the model will rely on its ability to learn relevant spatial and temporal patterns directly from the raw keypoint data.

3.7 Reliability and Validity

Reliability will be ensured through multiple measures:

1. Videos were recorded under consistent conditions (camera angle, lighting, background).
2. MoveNet Thunder provided consistent keypoint detection across frames, reducing human annotation variability.
3. Resampling to 50 frames ensured temporal consistency, reducing potential noise from variable video durations.
4. Mild rotations and scaling increased generalization and reduced overfitting, ensuring reliable performance across different camera perspectives.

Validity:

1. Controlled environment and consistent recording procedures reduced confounding variables.
2. The combination of keypoints, 1D-CNN, and LSTM layers directly measures squat posture and type, matching the intended construct.
3. While the dataset reflects controlled conditions, augmentation simulates minor variations in camera position and scale, improving generalization to real-world use.

3.8 Limitations

These limitations could affect the model's performance when applied to real-world videos outside the controlled dataset:

1. Videos will be recorded under standardized conditions. The model may perform less well in uncontrolled environments with different lighting or cluttered backgrounds.
2. Participants may not represent the full range of body types, potentially reducing generalization.
3. Resampling all sequences to 50 frames may obscure micro-variations in longer or shorter squats.
4. MoveNet Thunder, while accurate, may have errors in occluded or poorly lit frames, introducing noise into keypoint data.
5. Posture labels are categorical and may not capture the full spectrum of deviations from ideal form.

3.9 Summary

In summary, this study adopts a quantitative research design with the primary objective of developing and evaluating a system capable of accurately classifying squat types and identifying common posture errors. The dataset will consist of video recordings collected in a controlled environment under consistent conditions, with the author serving as the sole participant performing the exercises. In total, 12 videos will be recorded and segmented into approximately 600 shorter, clearly labeled clips.

Data preprocessing will include ensuring frame rate consistency, applying fixed-length resampling, removing audio, and reducing file sizes. To enhance model generalization, data augmentation techniques such as zooming, rotation, and translation will be applied to simulate realistic variations in user alignment, expanding the dataset to approximately 1,200 video recordings.

The pose estimation model to be used in this study is MoveNet Thunder, which is capable of detecting 2D joint locations across 17 keypoints. Using OpenCV, all recordings will be standardized to a fixed length of 50 frames for both training and testing, ensuring a consistent input tensor shape of $(50 \times 17 \times 2)$.

The extracted keypoints will serve as input to a temporal model combining a 1D convolutional neural network (1D-CNN) and a long short-term memory (LSTM) network, enabling the system to learn and classify movement patterns over time. The model will be trained to

classify squat styles (Back, Front, and Goblet) as well as posture variants (Correct, Insufficient Depth, Rounded Back, and Weight on Toes). The architecture will consist of six layers: a temporal convolutional layer, a pooling layer, a recurrent layer, a shared dense layer, and two task-specific output layers.

The model will output predefined templates based on the predicted squat style and posture. These templates will be refined through integration with Gemini, enabling the generation of natural, fluent, and actionable feedback for users.

The methodological choices are aligned with the research objectives of accurately classifying squat types, detecting posture deviations, and providing actionable feedback in a robust and reliable manner. While several limitations are anticipated, steps will be taken to reduce their impact through careful dataset design, augmentation, and qualitative verification.

Overall, the proposed methodology provides a solid foundation for developing and evaluating an automated squat assessment system capable of delivering accurate and user-friendly feedback.

4

Results/Analysis

4.1 Dataset Overview

4.1.1 After recording

The training videos were recorded to capture all three movement styles, with representative examples of styles shown in figure 4.1 and the amount of recording shown in table 4.1. During recording, a pole was used in place of an actual weighted barbell to ensure consistency and safety across trials. After recording, the footage was clipped using Microsoft Clipchamp. The processed videos were then exported and downloaded at a resolution of 480p, resulting in a final frame size of 540×540 pixels. The file sizes of the recorded videos ranged from approximately 400 KB to 1.1 MB.



Figure 4.1: Picture of all squat types: Front, Back and Goblet

Table 4.1: Recorded and clipped training data.

Back squat	Recordings	Front squat	Recordings	Goblet squat	Recordings
Correct	50	Correct	50	Correct	50
Insufficient depth	50	Insufficient depth	50	Insufficient depth	50
Rounded back	50	Rounded back	50	Rounded back	50
Weight on toes	50	Weight on toes	50	Weight on toes	50
Sum	200	Sum	200	Sum	200

The testing videos were recorded using two different participants. Participant 1 was a woman in her 30s who was shorter in height than the author featured in the testing videos. A female participant was intentionally selected to evaluate potential limitations of the model with respect to differences in body proportions. Participant 2 was a slimmer male in his late 50s who was the same height as the author. This participant was included to assess model performance across variations in age and body composition.

All testing videos were recorded using the same setup and procedure as the training videos. Each of the three movement styles was captured across four different postures. The recordings were subsequently trimmed using Microsoft Clipchamp and saved at a resolution of 1080p, resulting in a final frame size of 1080 × 1080 pixels. Images of the participants are shown in figure 4.2. The file sizes of the recorded videos ranged from approximately 950 KB to 1.2 MB.

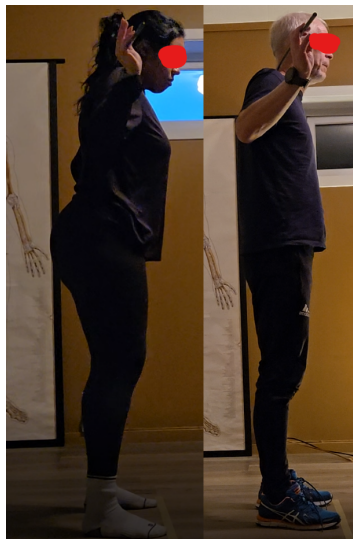


Figure 4.2: Picture of the two test participant

All participants, including the author in the training videos, wore black clothing. This choice was incidental, as participants were instructed prior to filming to wear tight-fitting clothing without stripes and in a color distinct from the background. However, the consistent use of black clothing may introduce bias into the model, and different clothing colors could potentially yield different model performance. Each participant recorded four videos for each

movement style and posture, resulting in a total of 96 test videos. See table 4.2. Although the original project proposal specified 48 test videos, both participants independently recorded 48 videos each. Consequently, all 96 videos were included in the final dataset. The dataset was manually labeled and organized using a folder hierarchy in Google Drive, as illustrated in figure 4.3

Table 4.2: Recorded and clipped test data.

Back squat	Recordings	Front squat	Recordings	Goblet squat	Recordings
Correct	8	Correct	8	Correct	8
Insufficient depth	8	Insufficient depth	8	Insufficient depth	8
Rounded back	8	Rounded back	8	Rounded back	8
Weight on toes	8	Weight on toes	8	Weight on toes	8
Sum	32	Sum	32	Sum	32

4.1.2 After Preprocessing

To ensure that the original recordings were preserved, separate folders were created 4.3.

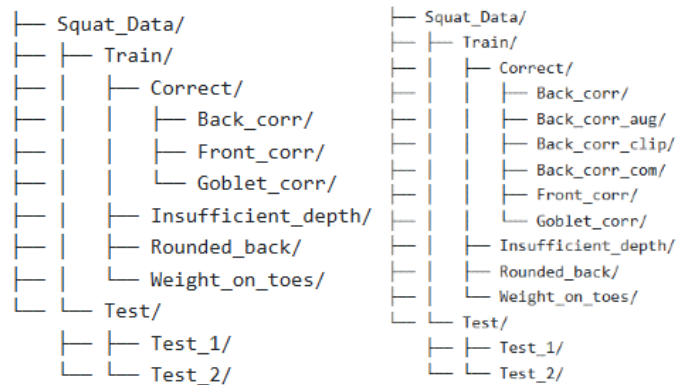


Figure 4.3: The first folder hierarchy: left. Added folders: right

The original clipped videos were stored in a designated “_clip” folder. The dataset was processed using Python 3 in Google Colab. FFmpeg was used to downscale the videos to a resolution of 480 × 270 pixels and to remove the audio tracks. The resulting processed videos were stored in the “_com” folder.

The compressed videos were then duplicated and augmented using OpenCV for Python, after which the augmented files were stored in the “_aug” folder. All videos from the “_com” and “_aug” folders were subsequently copied into the “_corr” folder. Each main folder contained 100 videos, resulting in a total of 1,200 test videos across the dataset. See table 4.3. The file sizes of the processed recordings ranged from 45 KB to 115 KB.

Table 4.3: Final dataset for training

Back squat	Recordings	Front squat	Recordings	Goblet squat	Recordings
Correct	100	Correct	100	Correct	100
Insufficient depth	100	Insufficient depth	100	Insufficient depth	100
Rounded back	100	Rounded back	100	Rounded back	100
Weight on toes	100	Weight on toes	100	Weight on toes	100
Sum	400	Sum	400	Sum	400

All video samples were standardized to a fixed temporal length of 50 frames. Each frame was represented by 17 anatomical keypoints extracted using the MoveNet Thunder pose estimation model. Consequently, each video was converted into a temporal sequence with a fixed shape of $(50 \times 17 \times 2)$, representing the x- and y-coordinates of each keypoint over time. The extracted keypoint data were saved as .npz files within a new folder hierarchy, as shown in figure 4.4.

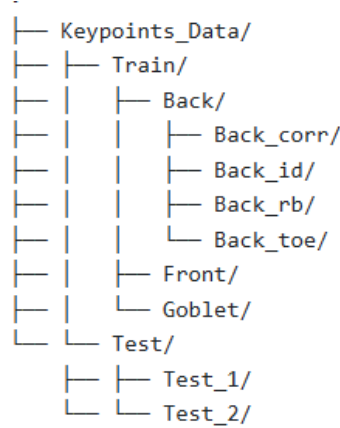


Figure 4.4: Keypoint folder hierarchy

4.2 Temporal Characteristics of the Pose-Based Representation

To verify the integrity of the extracted keypoint data, 10 random .npz files were selected and inspected to confirm their shape. This verification was performed using Python and the os library to randomly sample files from a directory and ensure that each array had the expected shape of $(50, 17, 2)$.

Further analysis of the extracted pose sequences indicated a high degree of temporal consistency across frames. Two random .npz files were selected from each of the folders “Back_corr,” “Goblet_rb,” and “Front_id” to assess temporal stability. As shown in 4.5, the temporal joint trajectories exhibit smooth and continuous variations in both horizontal and vertical coordinates across frames. No irregular jumps or frame-to-frame jitter were observed, indicating stable pose estimation performance across the six sampled sequences.

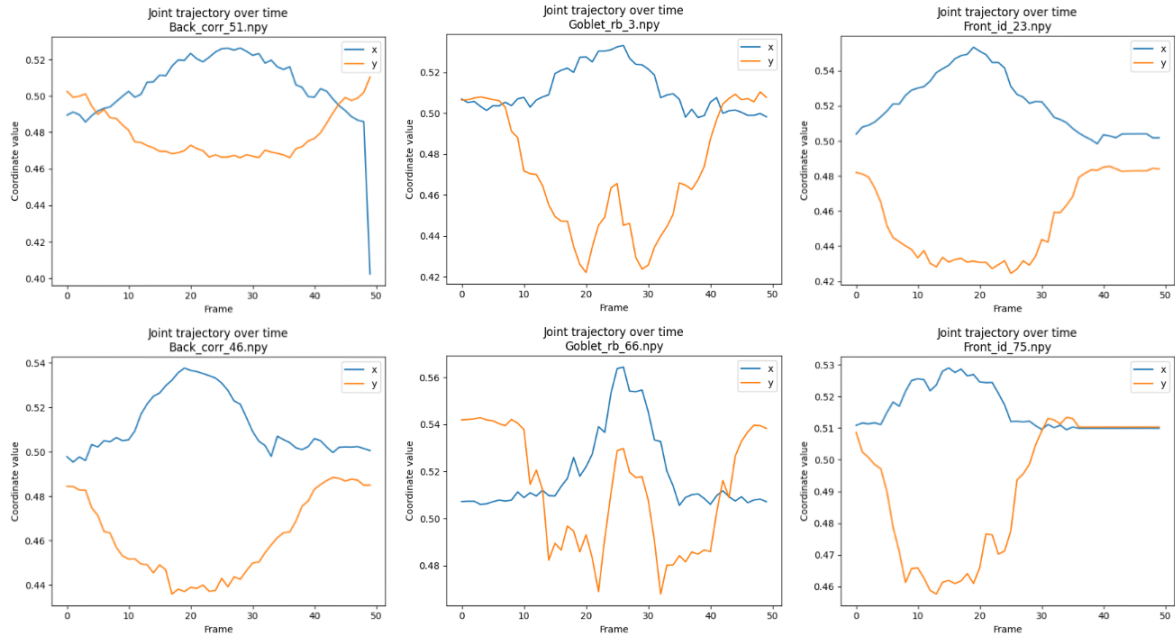


Figure 4.5: Temporal joint trajectories over 50 frames

Analysis of frame-to-frame keypoint displacement from the six sampled .npy files further supports these observations. The average temporal displacement was 0.0076 for correct back squat executions, 0.0109 for rounded-back goblet squat executions, and 0.0046 for insufficient-depth front squat executions. See table 4.4. These low values indicate a high degree of temporal stability across frames. The slightly higher displacement observed in rounded-back executions suggests increased upper-body movement rather than pose estimation noise.

Given the controlled recording setup, where all videos were captured using the same camera, location, and subject. The extracted keypoint coordinates remained highly consistent across samples. This is further supported by the spatial normalization results, where normalized keypoint coordinates ranged approximately between 0.33 and 0.70 across all samples.

While this consistency may simplify model learning, it may also limit the model's ability to generalize to more diverse real-world conditions.

Table 4.4: Temporal Stability and Spatial Normalization

	Back_corr	Goblet_rb	Front_id
Stability	0.00759	0.01088	0.00457
Spatial normalization	0.34–0.69	0.35–0.68	0.33–0.70

4.3 Comparison of 1D-CNN and 1D-CNN + LSTM Models

4.3.1 Training + posture classification

To evaluate the impact of explicit temporal modeling, the performance of a 1D convolutional neural network (1D-CNN) was compared with that of a hybrid CNN–long short-term memory network (LSTM) architecture. Both models were trained using identical data splits, input representations, and training hyperparameters, differing only in the inclusion of the LSTM layer. The 1D-CNN model achieved a training accuracy of 93.68% and a validation accuracy of 91.39%, indicating strong learning capacity and good generalization performance. The corresponding training and validation losses were 0.1980 and 0.2184, respectively, with a relatively small gap between training and validation metrics.

In contrast, the CNN–LSTM model achieved a lower training accuracy of 84.17% and a validation accuracy of 87.50%. The associated loss values were higher, with a training loss of 0.3401 and a validation loss of 0.3152. These results indicate that the inclusion of an LSTM layer did not improve classification performance for the given task. See table 4.5

Overall, the CNN-only model outperformed the CNN–LSTM architecture in both training and validation settings. This suggests that the temporal dependencies present in the extracted pose sequences were sufficiently captured by convolutional operations alone, and that explicit long-term temporal modeling did not provide additional benefit under the given experimental conditions.

Table 4.5: Training and validation metrics of training + posture

Model	Accuracy	Loss	Val_accuracy	Val_loss
CNN–LSTM	0.8417	0.3401	0.8750	0.3152
CNN	0.9368	0.1980	0.9139	0.2184

4.3.2 Training + style classification

To evaluate squat style classification, we did the same as we did with Training + posture classification. 1D convolutional neural network (CNN) and a hybrid CNN–LSTM model were trained using a 70/30 train/validation split. The CNN model achieved a training accuracy of 83.78% and a validation accuracy of 87.50%, indicating strong learning capacity and good generalization. In comparison, the CNN–LSTM model achieved a lower training accuracy of 69.99% and a validation accuracy of 78.61%. Overall classification accuracy was consistently higher for the CNN-only architecture. See table 4.6

These results suggest that explicit temporal modeling via an LSTM layer did not improve squat style classification performance and that local temporal features captured by convolutional layers were sufficient for this task.

Table 4.6: Training and validation metrics of training + Style

Model	Accuracy	Loss	Val_accuracy	Val_loss
CNN-LSTM	0.6999	0.07672	0.7861	0.3857
CNN	0.8378	0.4055	0.8750	0.3172

4.3.3 Test + Multi-Output classification

Squat Style Classification

A comparison of squat style classification performance between the CNN and CNN-LSTM multi-output models revealed no meaningful improvement from the inclusion of temporal memory. On the test set, both models achieved near chance-level accuracy, with the CNN achieving 29% accuracy and the CNN-LSTM achieving 30%. See table 4.7

Table 4.7: Classification performance per squat type

Class	Metric	CNN	CNN + LSTM
Back	Precision	0.00	0.00
	Recall	0.00	0.00
	F1-score	0.00	0.00
Front	Precision	0.29	0.30
	Recall	0.38	0.34
	F1-score	0.33	0.32
Goblet	Precision	0.29	0.31
	Recall	0.50	0.56
	F1-score	0.37	0.40
Overall accuracy		0.29	0.30

Per-class precision, recall, and F1-score metrics were comparable across models. The Back squat class was not correctly identified by either model, while Front and Goblet squats exhibited moderate recall with substantial confusion between the two classes. Although the CNN-LSTM model achieved a slightly higher recall for the Goblet class, this improvement did not translate into a significant overall performance gain. The similarity of the confusion matrices (figure 4.6) across both architectures indicates that explicit temporal modeling did not alter the underlying error patterns for squat style classification.

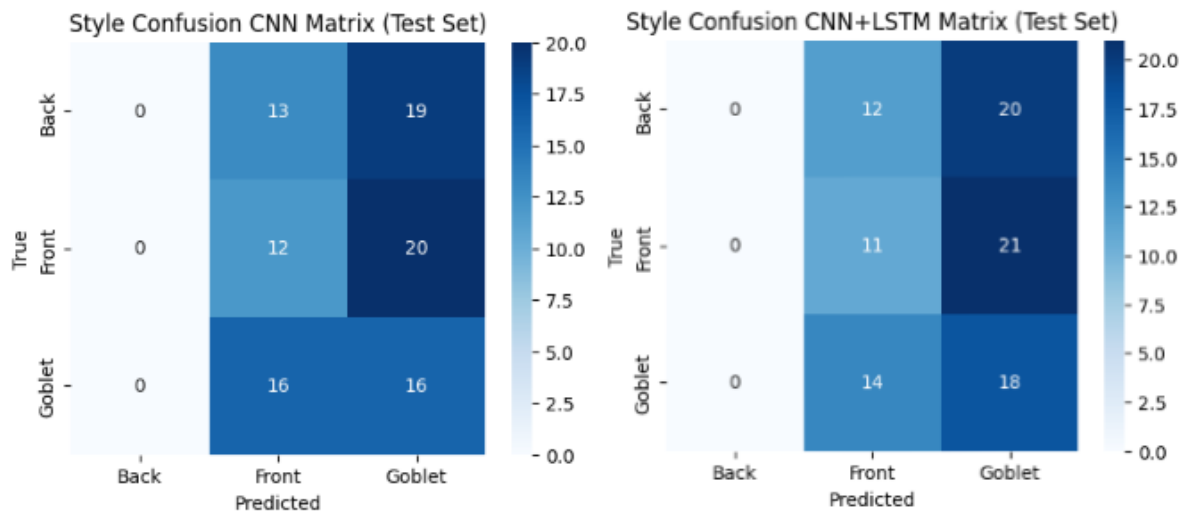


Figure 4.6: The squat style execution form classification confusion matrices illustrate the performance of the CNN model (on the left) and the CNN+LSTM model (on the right) when tested on the test dataset.

Squat posture Classification

A comparison of posture classification performance between the CNN and CNN–LSTM multi-output models revealed complementary strengths. Both models scored around the same at accuracy, CNN at 66% and CNN–LSTM at 69%. Also both models demonstrated strong performance for the Insufficient Depth and Rounded Back classes. See table 4.7. The CNN–LSTM model achieved perfect classification for the Rounded Back class and substantially improved recall for the Weight-on-Toes class, indicating improved sensitivity to temporally consistent posture deviations. See table 4.8

Table 4.8: Classification performance per posture class

Posture class	Metric	CNN	CNN + LSTM
Correct	Precision	0.50	0.00
	Recall	0.04	0.00
	F1-score	0.08	0.00
Insufficient depth	Precision	0.71	0.95
	Recall	1.00	0.83
	F1-score	0.83	0.89
Rounded back	Precision	1.00	1.00
	Recall	0.92	1.00
	F1-score	0.96	1.00
Weight on toes	Precision	0.42	0.46
	Recall	0.67	0.92
	F1-score	0.52	0.61
Overall accuracy		0.66	0.69

In contrast, the CNN model exhibited slightly better performance for Correct executions, although recall for this class remained low overall. The CNN+LSTM model failed to identify Correct executions on the test set, suggesting a more conservative classification behavior that favors error detection. Overall, the inclusion of an LSTM layer improved the detection of dynamic posture errors but reduced sensitivity to correct movement patterns.

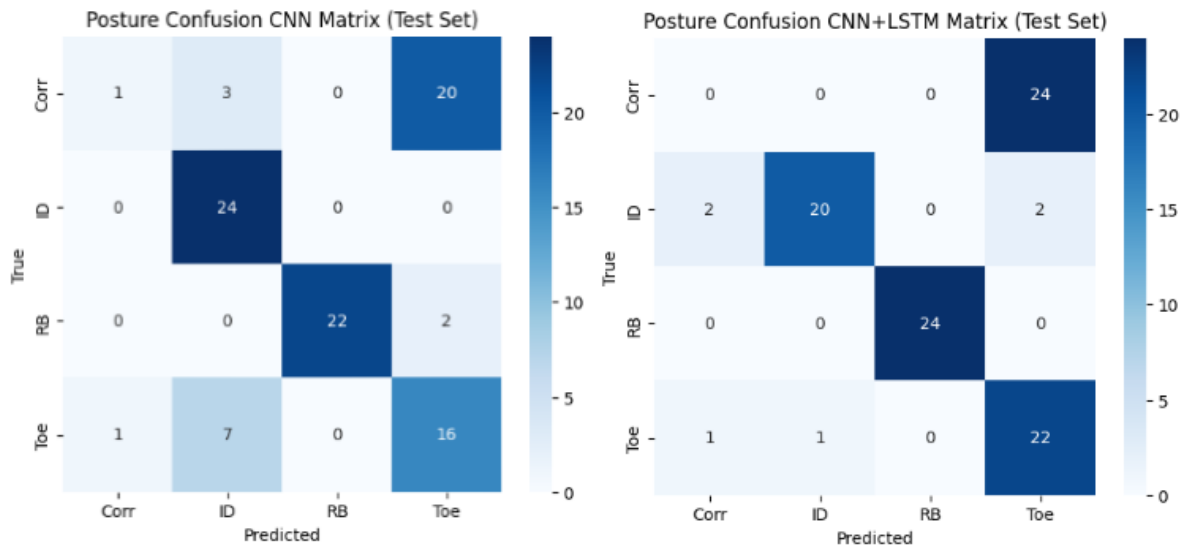


Figure 4.7: The squat posture execution form classification confusion matrices illustrate the performance of the CNN model (on the left) and the CNN+LSTM model (on the right) when tested on the test dataset.

4.4 End-to-End System Evaluation

To evaluate the full end-to-end pipeline, a test video from Participant 2 performing a goblet squat with insufficient depth was processed through the complete system, including pose estimation, classification, and automated feedback generation. The posture classification component correctly identified the execution error as Insufficient Depth. However, the squat style was not classified as Goblet, which is consistent with the previously observed limitations of the style classification model on unseen participants.

Despite the incorrect style, the generated feedback remained meaningful and contextually appropriate, as the posture classification result was correctly identified. The Gemini-based rephrasing component successfully transformed the predefined corrective template into a natural-language coaching message while preserving the original corrective intent. This demonstrates that the feedback generation component operates independently of style classification accuracy and remains effective as long as posture-related errors are correctly detected. The chosen template for posture and style can be seen on table 4.10 and 4.9

Table 4.9: Style feedback templates

Style class	Feedback message
Back	You are performing a back squat.
Front	You are performing a front squat.
Goblet	You are performing a goblet squat.

Table 4.10: Posture feedback templates

Posture class	Feedback message
Correct	Your squat technique is correct.
Insufficient depth	You are not reaching sufficient squat depth.
Rounded back	Your lower back is rounding. Keep a neutral spine.
Weight on toes	Your weight is shifting toward your toes. Keep pressure on your heels.

Figures 4.8 and 4.9 illustrate the differences in responses generated by Gemini. The response produced in the Colab pipeline appeared more motivating compared to the one generated within the artifact. Additionally, Gemini produced slightly different responses each time the pipeline was executed.

Although the outputs consistently followed the structure of the provided template, the level of motivational language varied. In most cases, the feedback included both corrective and motivational elements; however, in some instances, it was limited to corrective feedback only.

Attempts to enforce more detailed prompting rules resulted in less coherent responses. Consequently, the most effective approach was to use a simple instruction: “Rewrite this into one short sentence (maximum 12 words). No explanations.”

```

final_feedback = rephrase_feedback_with_gemini(template_text)

print("\nFinal Gemini feedback:")
print(final_feedback)

...
Final Gemini feedback:
"Great job with those front squats; just try to get a bit more depth by lowering your hips further."

```

Figure 4.8: Collab:Feedback from Gemini

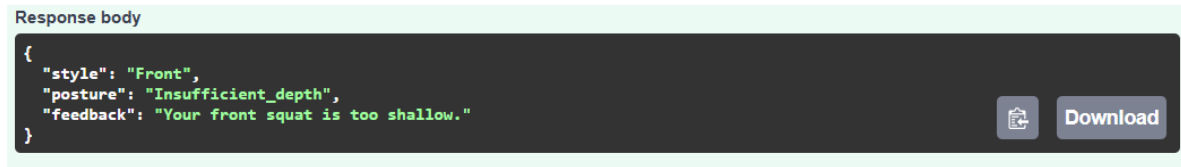


Figure 4.9: Artifact:Feedback from Gemini

4.5 Experimental Results and Discussion

Experiment	Why	Input	Expectation	Result
Experiment 1	The multi-output model showed high training accuracy but poor test performance. Therefore, a hierarchical model was proposed.	Training dataset with multiple target labels. Evaluated on a separate test set.	The model should correctly predict all target labels with high accuracy.	High training accuracy but poor test performance.
Experiment 2	To investigate the poor model performance, temporal pose data was examined.	Test set keypoint data.	Sudden jumps, missing frames, or abnormal values that might explain the poor performance.	The data was properly scaled, contained no abnormal or out-of-range values, and showed smooth and continuous motion across the 50 frames.
Experiment 3	To check whether the model has difficulty generalizing across different participants.	1080 training samples with multiple target labels. 120 samples used as a test set.	The model should correctly predict all target labels with high accuracy.	The model performed very well and predicted almost all target labels with high accuracy.
Experiment 4	To evaluate whether the model exhibits bias toward gender, body type, or individual-specific patterns when applied to unseen participants. to unseen participants.	Separate test datasets for each participant.	Overall poor performance was expected, with slightly higher accuracy anticipated for Participant 2 (male).	The models demonstrated poor performance for both participants, with no clear evidence of bias toward gender or body type.

Table 4.11: Summary of experiments

4.5.1 Experiment 1

The multi-output classification model did not perform as expected when evaluated on the external test dataset. Based on these results, a new approach was proposed, the development of a hierarchical model architecture.

In this architecture, the test data is first processed by a style classification model, which previously demonstrated strong performance during training. Depending on the predicted squat style, the sample is then routed to one of three separate posture classification models. Each of these models is trained specifically to classify posture within a single squat style.

For this hierarchical approach, the 1D-CNN model without the LSTM layer was selected. The LSTM component was removed because it increased the overall model complexity without providing measurable performance improvements in earlier experiments.

To further improve performance, the number of training epochs for the style classification model was increased from 30 to 100. See table 4.12. Since this model serves as the first stage in the hierarchical architecture, improving its performance was considered particularly important. The results showed a 9.43% increase in training accuracy and a 7.78% increase in validation accuracy.

Model	Accuracy	Loss	Val_Accuracy	Val_Loss
30 Epoch	0.8378	0.4055	0.8750	0.3172
100 Epoch	0.9321	0.1798	0.9528	0.1090

Table 4.12: Experimental style epoch results

Before implementing the full hierarchical model, the improved style classification model was evaluated on the external test dataset. See table 4.13. However, the model achieved an accuracy of only 38%, with the majority of test samples predicted as front squat regardless of the true class. The back squat class was once again not identified, and overall performance was worse than that of the multi-output model, as shown in Figure 4.10.

Class	Precision	Recall	F1-score
Back	0.00	0.00	0.00
Front	0.34	0.94	0.50
Goblet	0.75	0.19	0.30
Accuracy	0.38		

Table 4.13: Experimental style classification on testset

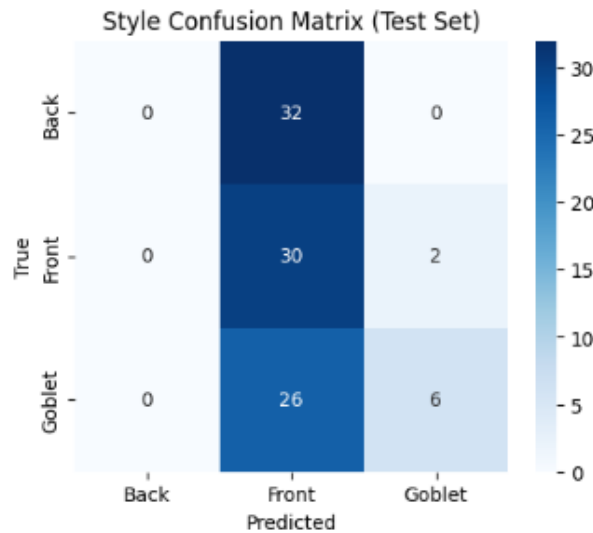


Figure 4.10: Experimental CNN squat style classification model on test set.

4.5.2 Experiment 2

The new model also showed low accuracy when evaluated on the external test dataset. When the style classification model, intended for the hierarchical approach, was tested independently, its performance remained poor. This raised the question of whether the issue stemmed from problems in the test dataset or from limitations in the model when applied to unseen participants.

To better understand the reduced model performance, the temporal pose data from the test dataset was analysed. The objective was to verify that the keypoint sequences were stable and correctly processed.

Several samples were randomly selected from both test participants and from different squat styles. The coordinate values were examined by analysing their minimum and maximum values. The results showed that the values remained within a consistent normalized range, indicating that the pose data was properly scaled and did not contain abnormal or out-of-range values. See table 4.14.

Participant	Test Sample	Squat Style	Min Value	Max Value
Participant 1	Test_back_corr_1	Back	0.30422533	0.8722489
Participant 1	Test_goblet_rb_3	Goblet	0.28700447	0.8816568
Participant 1	Test_front_toe_4	Front	0.31869876	0.87626964
Participant 2	Test2_back_corr_4	Back	0.17443776	0.7826484
Participant 2	Test2_goblet_rb_1	Goblet	0.17197861	0.78841037
Participant 2	Test2_front_toe_1	Front	0.16371134	0.7854184

Table 4.14: Minimum and maximum values for each test sample

In addition, joint trajectories were visualized to observe the movement patterns over time. The resulting plots showed smooth and continuous motion across the 50 frames, corresponding to the expected phases of a squat movement, including descent and ascent. No sudden jumps, missing frames, or unstable tracking were observed. See figure 4.11 and 4.12.

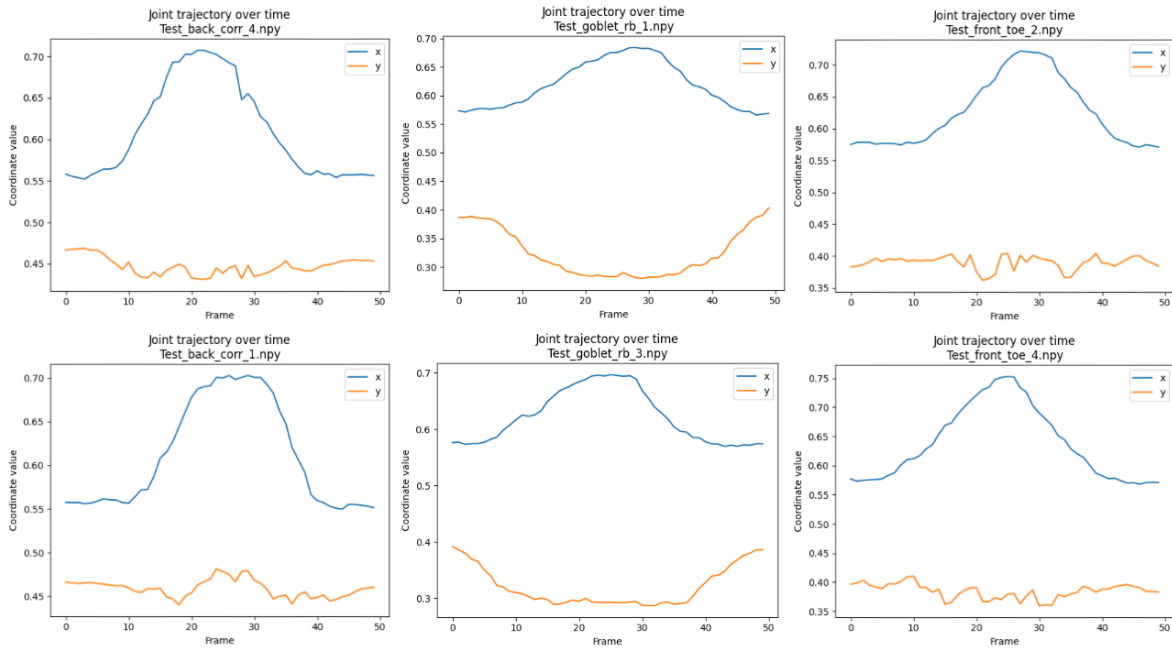


Figure 4.11: Temporal joint trajectories of participant 1

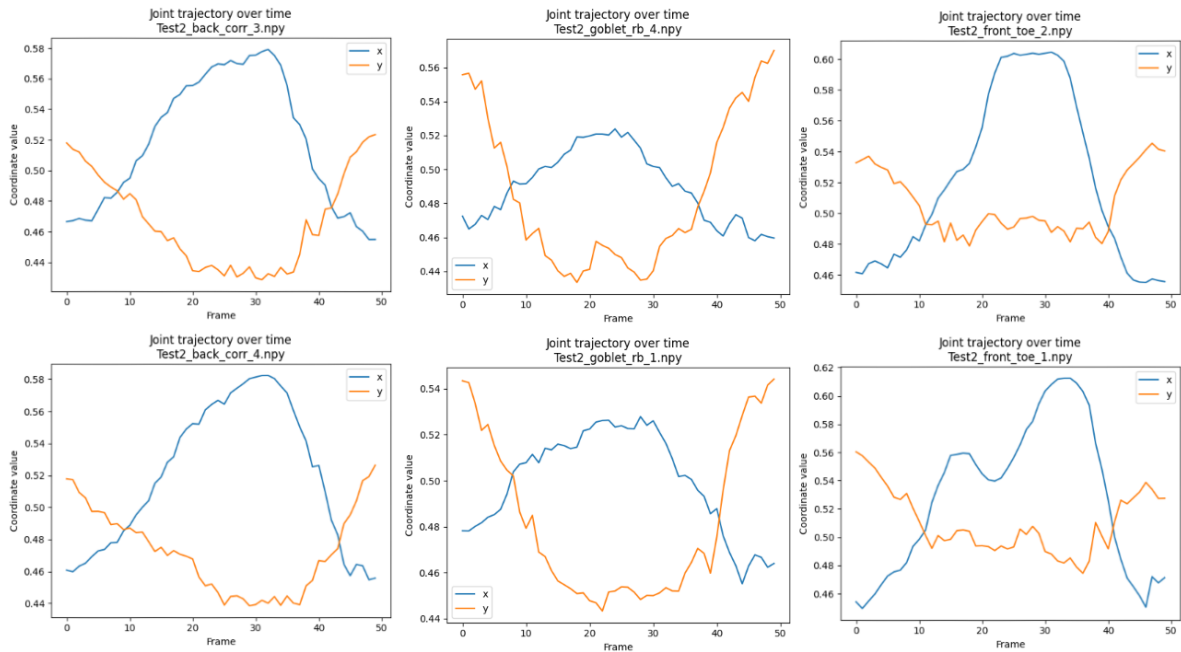


Figure 4.12: Temporal joint trajectories of participant 2

4.5.3 Experiment 3

Since no issues were found in the temporal pose data, the low accuracy on the external test dataset is unlikely to be caused by errors in the keypoint sequences. Instead, the reduced performance may be related to how the model handles data from new participants.

To further investigate this, the same model architecture was rebuilt and trained using only the

original training dataset. From the 1200 available keypoint recordings, 1080 samples were used for training and validation with a 70/30 split. The remaining 120 samples were reserved for testing. These test samples were selected by taking 10 recordings from each squat style and posture category. This setup allowed the model to be evaluated on unseen samples while still using data from the same participant.

Training of the new model showed improved performance in both training and validation compared to the previous multi-output model. This improvement is likely due to the increase in the number of training epochs from 30 to 100, as earlier experiments indicated that a higher number of epochs enhanced model performance. For posture classification, the model achieved a training accuracy of 99.55% and a validation accuracy of 98.15%, with a loss below 8% in both cases. Style classification also showed strong performance, with training and validation accuracies of 91.53% and 93.52%, respectively, and a loss below 23%. See table 4.15.

Metric	Training	Validation
Posture		
Accuracy	0.9955	0.9815
Loss	0.0306	0.0748
Style		
Accuracy	0.9153	0.9352
Loss	0.2241	0.1930
Overall		
Loss	0.2545	0.2568

Table 4.15: Experimental 3: Training and validation metrics for posture and style

The retrained model shows a substantial improvement compared to the original model that was evaluated on data from new participants.

In contrast, the initial experiment showed significantly lower performance on the external test dataset. The style classification model achieved an accuracy of only 29% and failed completely to recognize back squats (precision and recall of 0.00). Most predictions were instead classified as front or goblet squats, indicating that the model struggled to generalize style characteristics when applied to participants not present in the training data.

Similarly, although the posture model achieved a moderate overall accuracy of 66% in the evaluation of the first model, performance varied significantly across classes. Correct posture was rarely detected (recall of 0.04), whereas other classes, such as rounded back and insufficient depth, were classified with much higher accuracy.

In contrast, the retrained model achieved significantly higher performance when both the training and test data originated from the same participant distribution. The style classification accuracy increased to 92%, with perfect classification of back squats and only minor confusion between front and goblet squats. This is in direct contrast to the model's performance

on unseen participants. The posture classification model also improved considerably, reaching an accuracy of 97%, with near-perfect precision and recall across all posture classes. Notably, the correct squat class, which previously showed very low performance, achieved a recall of 0.97 with 100% precision. See table 4.16, 4.17 and figure 4.13

Style	Precision	Recall	F1-score
Back	1.00	1.00	1.00
Front	0.86	0.90	0.88
Goblet	0.89	0.85	0.87
Accuracy	0.92		

Table 4.16: Experimental 3: Style classification

Table 4.17: Experimental 3: Classification performance per posture class

Posture class	Metric	Model
Correct	Precision	1.00
	Recall	0.97
	F1-score	0.98
Insufficient depth	Precision	0.91
	Recall	1.00
	F1-score	0.95
Rounded back	Precision	1.00
	Recall	1.00
	F1-score	1.00
Weight on toes	Precision	1.00
	Recall	0.93
	F1-score	0.97
Overall accuracy		0.97

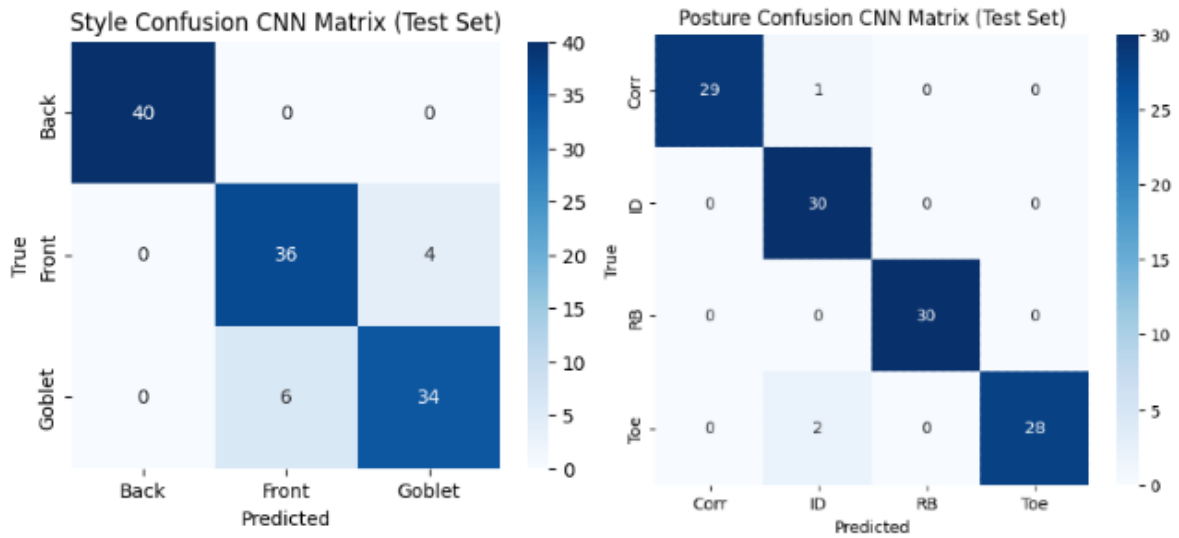


Figure 4.13: Experimental 3: Posture and style confusion CNN Matrix on new test dataset.

This comparison suggests that the model architecture is capable of learning meaningful features from the keypoint data. The large performance difference between the two experiments indicates that the primary limitation lies in the model's ability to generalize across different participants, rather than in the CNN architecture itself.

The results show that the model can perform very well when evaluated on data from the same individual it was trained on. This further supports the theory that the model is learning participant-specific patterns rather than generalizable features.

These findings also suggest that performance on unseen individuals could be improved by increasing the diversity of the training dataset, particularly by including a wider range of participants.

4.5.4 Experiment 4

Due to limitations in the model's ability to generalize across different participants, an additional experiment was conducted. Given that data from two participants: a female subject and an elderly male subject were available, a test was performed to evaluate whether the model exhibited bias toward gender, body type, or individual-specific patterns when applied to unseen participants.

To perform this evaluation, the same trained model used in the artifact, as well as a CNN-only baseline model, were employed. The evaluation was conducted separately using the test set from Participant 1 (female) and Participant 2 (elderly male).

As expected, the multi-output classification model did not perform well. The accuracy of squat style classification was 27% for Participant 1 and 33% for Participant 2. The back squat was

not identified in either participant, which is consistent with previous results obtained when both participants were included in the test set. See table 4.18.

An unexpected pattern emerged: the model identified only the front squat for Participant 1, while for Participant 2 it identified only the goblet squat, achieving perfect recall for that class. This suggests that the model may be learning participant-specific features rather than generalizable movement patterns. See figure 4.14 for confusion matrix.

Table 4.18: CNN-only: Style Classification Comparison of Participants

Class	Participant 1			Participant 2		
	Precision	Recall	F1	Precision	Recall	F1
Back	0.00	0.00	0.00	0.00	0.00	0.00
Front	0.29	0.75	0.42	0.00	0.00	0.00
Goblet	0.00	0.00	0.00	0.33	1.00	0.50
Accuracy	0.27			0.33		

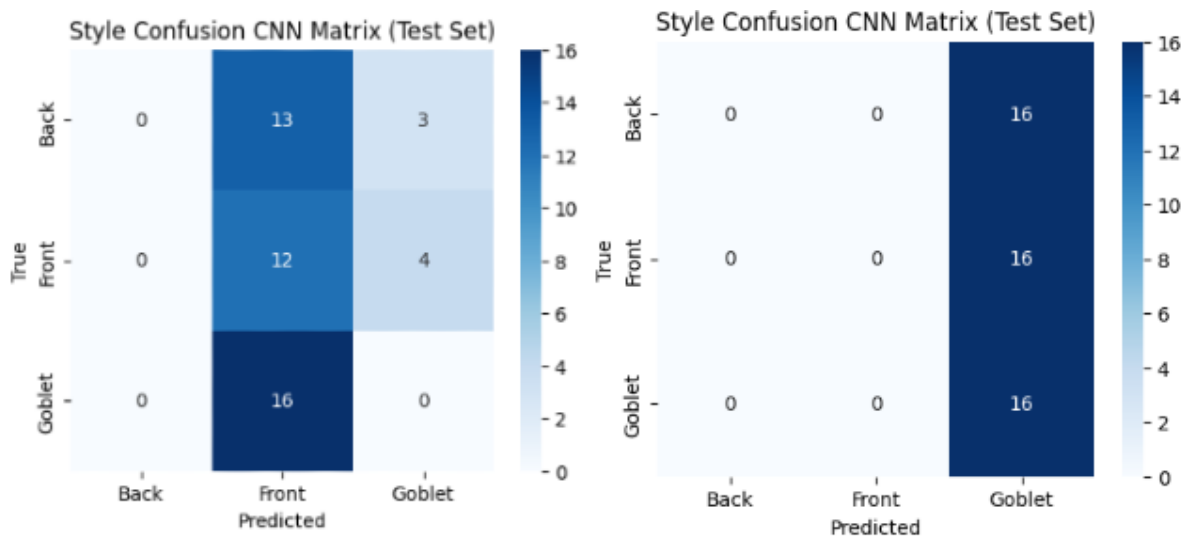


Figure 4.14: Experimental 4: Style Classification Comparison of Participants. Participant 1 on the left and Participant 2 on the right.

Examining the posture classification results in Table 4.19, both participants achieved identical precision, recall, and F1-scores for the rounded back class. Participant 1 achieved a higher overall accuracy 73% compared to Participant 2 58%.

Both participants achieved perfect recall for the insufficient depth class; however, Participant 1 demonstrated higher precision and F1-score overall. Notably, both participants struggled to correctly classify the correct squat class, with Participant 2 achieving no correct classifications recall = 0.00, and Participant 1 performing only marginally better with recall of 0.08.

Additionally, Participant 1 outperformed Participant 2 in detecting the weight on toes posture. Overall, the model appears to perform better on distinct error patterns than on identifying

correct execution, further indicating limitations in generalization across participants. See figure 4.15 for confusion matrix.

Table 4.19: CNN-only: Posture Comparison of Participants

Posture class	Metric	Participant 1	Participant 2
Correct	Precision	0.50	0.00
	Recall	0.08	0.00
	F1-score	0.14	0.00
Insufficient depth	Precision	0.92	0.57
	Recall	1.00	1.00
	F1-score	0.96	0.73
Rounded back	Precision	1.00	1.00
	Recall	0.92	0.92
	F1-score	0.96	0.96
Weight on toes	Precision	0.50	0.31
	Recall	0.92	0.42
	F1-score	0.65	0.36
Overall accuracy		0.73	0.58

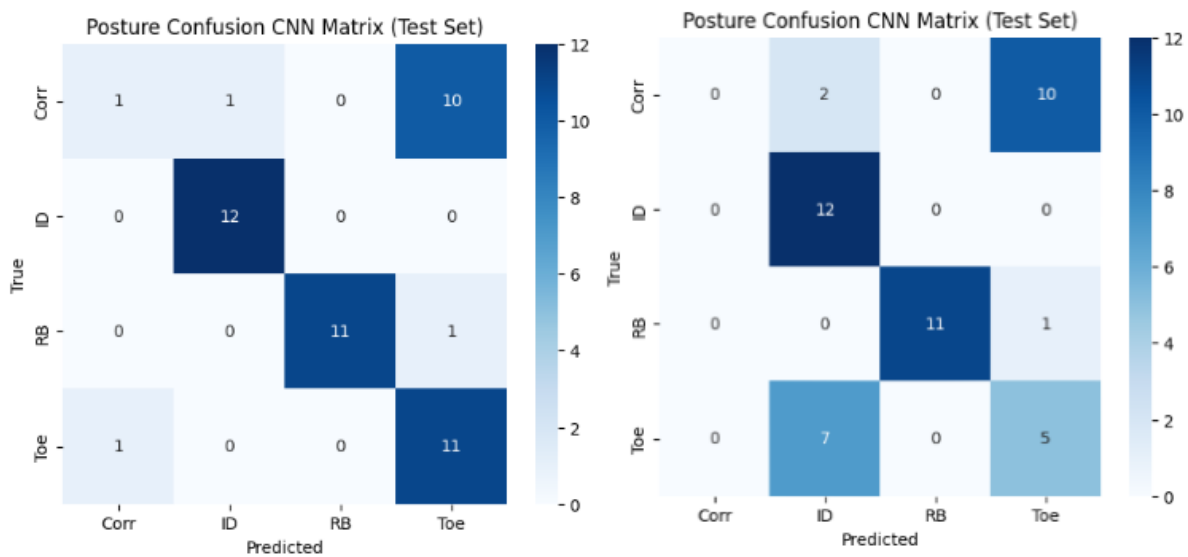


Figure 4.15: Experimental 4: Posture Classification Comparison of Participants. Participant 1 on the left and Participant 2 on the right.

When the same experiment was conducted using the main CNN–long short-term memory network (LSTM) model, the results were largely as expected, with one notable difference. As shown in Table 4.20, the back squat was still not correctly classified in this model. The overall accuracy remained similar to that of the CNN-only model.

However, unlike the CNN-only model, both participants were able to correctly classify instances of both front and goblet squats. This indicates that the inclusion of the LSTM im-

proved the model's ability to generalize squat style classification across unseen participants. Nevertheless, the overall performance remains limited. See figure 4.16 for confusion matrix.

Table 4.20: CNN-LSTM: Style Classification Comparison of Participants

Class	Participant 1			Participant 2		
	Precision	Recall	F1	Precision	Recall	F1
Back	0.00	0.00	0.00	0.00	0.00	0.00
Front	0.24	0.31	0.27	0.38	0.38	0.38
Goblet	0.30	0.50	0.37	0.31	0.62	0.42
Accuracy	0.27			0.33		

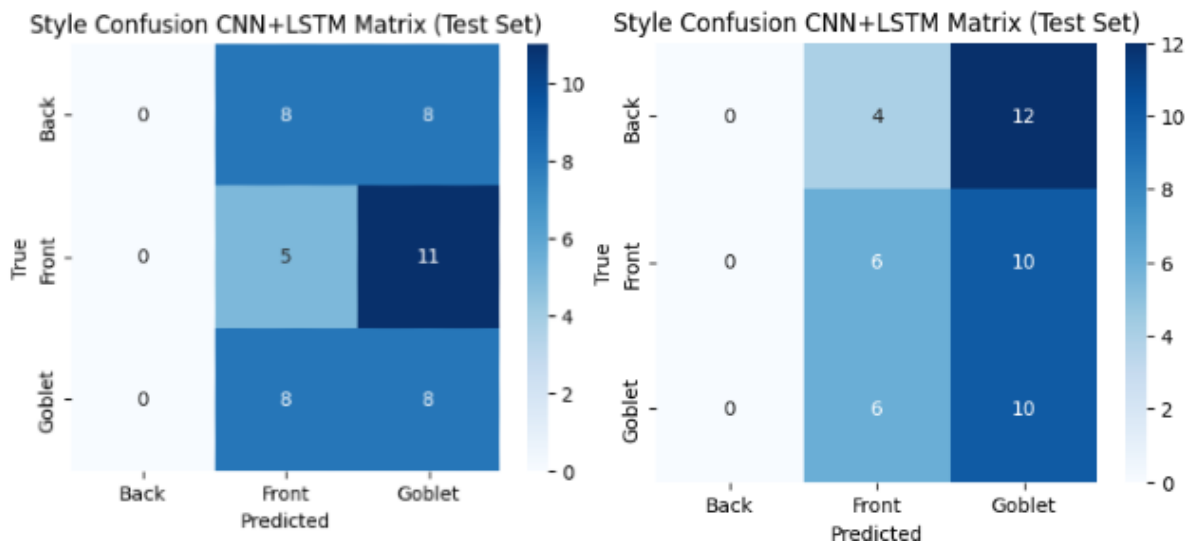


Figure 4.16: Experimental 4: Style confusion matrix comparison of participants (LSTM). Participant 1 on the left and Participant 2 on the right.

Examining the posture classification results in table 4.21, neither participant was able to correctly classify instances of the correct squat class. Participant 2 maintained perfect recall for the insufficient depth class, consistent with the CNN-only model, and achieved notably higher precision and F1-score when using the CNN-long short-term memory network (LSTM). In contrast, Participant 1 showed a decrease in performance for the same class, although precision remained perfect.

Both participants achieved perfect scores for the rounded back class. In the weight on toes category, Participant 1 showed similar performance to the CNN-only model, whereas Participant 2 demonstrated an improvement, achieving a recall of 0.92 along with slightly higher precision and F1-score.

Overall, the model achieved a lower accuracy for Participant 1: 65% and a higher accuracy for Participant 2: 73%. See figure 4.17 for confusion matrix

Table 4.21: CNN-LSTM: Posture Comparison of Participants

Posture class	Metric	Participant 1	Participant 2
Correct	Precision	0.00	0.00
	Recall	0.00	0.00
	F1-score	0.00	0.00
Insufficient depth	Precision	1.00	0.92
	Recall	0.67	1.00
	F1-score	0.80	0.96
Rounded back	Precision	1.00	1.00
	Recall	1.00	1.00
	F1-score	1.00	1.00
Weight on toes	Precision	0.44	0.48
	Recall	0.92	0.92
	F1-score	0.59	0.63
Overall accuracy		0.65	0.73

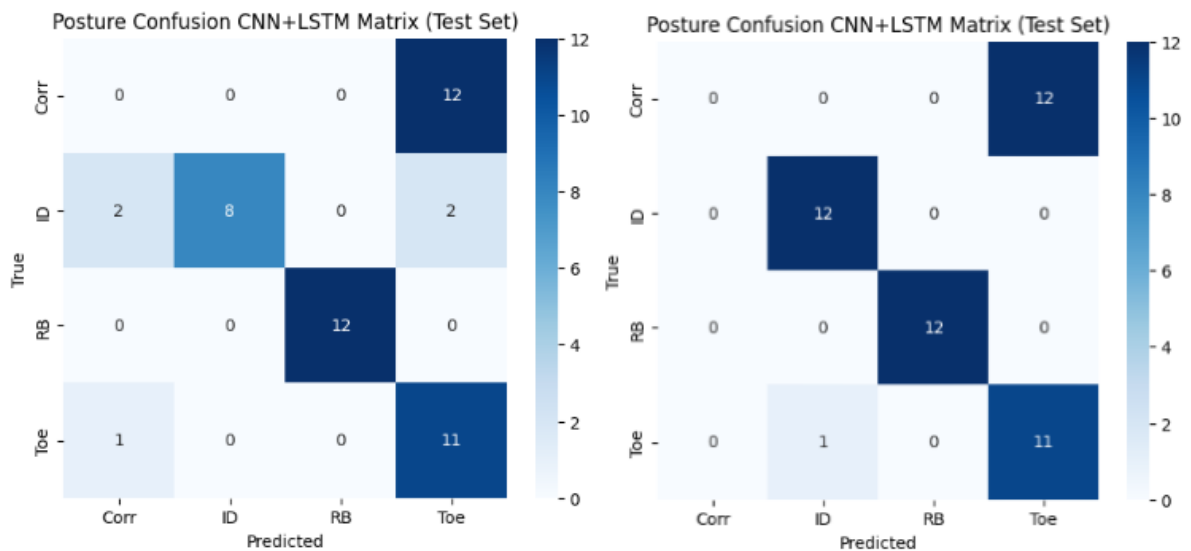


Figure 4.17: Experimental 4: Posture confusion matrix comparison of participants(LSTM). Participant 1 on the left and Participant 2 on the right.

The results show that the model tends to learn patterns specific to each individual rather than general patterns that work across different people. Its performance varied a lot between the two participants, temporal modeling improved results for one participant but made them worse for the other. Because only two participants were used, it is not possible to determine whether this bias is related to gender or body type. However, the results clearly show that the main limitation of the model is its poor ability to generalize to new participants.

4.6 Summary

This study evaluated a pose-based squat analysis system that integrates MoveNet pose estimation, deep learning-based classification, and large language model-driven feedback generation to analyze squat technique. The results demonstrate that temporal keypoint representations are more effective for posture error detection, while squat style classification remains more challenging.

The analysis of the extracted keypoint training data showed no irregular jumps or frame-to-frame jitter, indicating stable pose estimation performance across the six randomly selected samples. This was further supported by the low stability values and the consistent range of normalized keypoint coordinates, indicating a high degree of temporal stability across frames.

Across both training and validation experiments, the 1D-CNN consistently outperformed the CNN-long short-term memory network (LSTM) model in terms of accuracy. This indicates that the inclusion of an LSTM layer increased model complexity without providing performance benefits for squat style and posture classification.

Squat style classification (Back, Front, Goblet) showed strong performance on training and validation data but failed to generalize to unseen participants in the external test dataset. Both CNN and CNN-LSTM architectures produced similar results, with performance close to random guessing in subject-independent evaluations and noticeable prediction biases toward certain classes. Both models achieved strong performance in posture classification, particularly for detecting insufficient depth and rounded back errors. The CNN-long short-term memory network (LSTM) model achieved perfect performance on the rounded back class. These error types showed high precision and recall, indicating that pose-based features can reliably capture deviations from correct movement patterns during the squat.

However, correct squat executions were frequently misclassified as posture errors. The CNN model achieved a precision of 0.50 for the correct class, but recall and F1-score remained below 0.1, indicating poor detection. The CNN-LSTM model failed to identify any correct executions in the test set, suggesting a more conservative classification behaviour that favours error detection. Overall, both models appear to be more sensitive to detecting deviations from ideal movement patterns than accurately identifying correct technique.

Comparisons between CNN and CNN-LSTM architectures showed that the addition of the LSTM layer provided limited and task-dependent benefits relative to the increased model complexity. An end-to-end system evaluation on unseen test videos further confirmed these findings, where posture-related errors were correctly identified even when squat style was occasionally misclassified. Overall, the results indicate that pose-based temporal modelling is effective for posture error detection and automated feedback generation, while squat style recognition remains difficult to generalize across different individuals. To investigate why the

models showed strong performance on training and validation data but failed to generalize to unseen participants, additional experiments were conducted.

The first experiment explored replacing the multi-output model with a hierarchical approach. This involved using four separate models: one for squat style classification and three models specifically trained for posture classification. The style classification model was trained for 100 epochs instead of 30 in an attempt to improve accuracy. However, testing with a new participant showed that the model still performed poorly. Most test samples were predicted as front squat, regardless of the true squat style.

The second experiment analysis the test data keypoint to see if the keypoint sequences were stable and correctly processed. They results showed stability, no irregular jumps or frame-to-frame jitter.

The third experiment aimed to determine whether the low accuracy on the external test dataset was caused by the introduction of new participants. To investigate this, a new CNN-only model was trained on the same dataset, excluding 120 samples (10 from each category), which were instead used as a test set. The overall accuracy increased to over 90% for both squat style and posture classification. This suggests that the model architecture is capable of learning meaningful features from the keypoint data.

The final experiment aimed to determine whether there were performance differences between the two participants. As the participants differed in gender and age (one female and one elderly male), the experiment evaluated whether the model exhibited bias toward gender, body type, or individual-specific patterns when applied to unseen participants.

To assess this, both the CNN-only and CNN-long short-term memory network (LSTM) models were evaluated separately on each participant. The results showed significant variation in performance between the two participants. Temporal modeling improved results for one participant but degraded performance for the other. However, overall accuracy remained low for both participants.

4.7 Discussion

The results of this study highlight several important findings regarding the use of pose-based models for squat analysis. While the system demonstrated strong performance in controlled training and validation settings, its ability to generalize to unseen participants was limited. To better understand these outcomes, several aspects of the model design were examined, including the role of temporal modeling.

Temporal modeling is commonly used in human movement analysis to capture how motion evolves over time. In this study, a long short-term memory network (LSTM) was introduced to learn temporal dependencies in squat sequences. However, the results did not support the

expected performance improvements. The CNN-only model consistently outperformed the CNN-LSTM model in both squat style and posture classification during training and validation. This suggests that the inclusion of the LSTM layer increased model complexity without providing additional benefit.

One possible explanation for the limited benefit of the LSTM is the simplicity of the input data. The model was trained on keypoint sequences with a fixed length of 50 frames, which may be too short for temporal modeling to provide a significant advantage. Additionally, the movement patterns may already be sufficiently captured by the pose estimation model prior to classification, making the role of the LSTM redundant.

If the training data had included greater variation in subjects or less controlled recording conditions, such as differences in camera distance, lighting, clothing, and background, the LSTM may have had a greater impact on model performance, particularly in handling increased variability and noise. Additionally, the squat exercise is a repetitive and structured movement, which may reduce the importance of temporal modeling, as the key movement patterns can be captured from individual poses.

In the multi-output model, when the external test dataset was introduced, temporal modeling showed some improvement in sensitivity to certain temporally consistent posture deviations, such as a rounded back. However, overall performance remained nearly identical to that of the CNN-only model.

The results of the multi-output model also raise an important question as to why posture classification performed significantly better than squat style classification in both the CNN-only and CNN-LSTM models. One possible explanation is the clear definition of posture errors. For example, a rounded back is characterized by a forward head and downward gaze, while insufficient depth has a relatively clear threshold between correct and incorrect execution. These deviations are easier to detect compared to differences in squat style.

In contrast, the distinction between squat styles (back, front, and goblet) is more subtle. While goblet squats are more visually distinct due to hand positioning, back and front squats share many similar joint configurations, with differences primarily in upper-body positioning. As a result, these variations are more difficult to capture using only 2D keypoint data.

Overall, posture errors are easier to detect because they represent clear deviations from ideal movement, whereas squat styles require finer distinctions that are harder to capture using pose-based features. This suggests that not all classification tasks are equally learnable, and that posture error detection is more robust than squat style classification in pose-based models.

Despite strong performance during training and validation, the models failed when applied to unseen participants. This raises the question of why the models failed to generalize. Both the training and test datasets were collected under controlled conditions, meaning the

performance drop cannot be attributed to environmental differences. The same setup and camera were used, the primary difference was the individual performing the movement.

Although the training videos were downscaled to 256×256 resolution before training and the test videos used the original 1080p resolution, this difference is unlikely to explain the performance gap. Higher resolution should, if anything, improve keypoint detection, and the pose estimation model is designed to be robust to variations in resolution.

Pose estimation models typically normalize input and extract relative joint positions, making them less sensitive to differences in image resolution. Therefore, the observed performance drop is more likely due to differences between participants rather than differences in recording conditions, as resolution has a limited impact on the extracted keypoints.

An argument could be made that differences in body proportions contributed to the performance drop. However, the author and Participant 2 were of similar height, although Participant 2 was slightly leaner, while Participant 1 was shorter. Despite these differences, both participants showed poor performance when evaluated by the model.

The difference in gender between Participant 1 (female) and Participant 2 (male) did not result in a significant difference in model performance. However, this does not provide sufficient evidence to rule out gender-based bias, as the sample size is too small to draw such a conclusion. Overall, these observations suggest that body proportions alone are unlikely to explain the poor performance.

Another possible explanation is that the keypoints in the test data were corrupted. However, this is unlikely, as Experiment 2 demonstrated stable keypoint sequences with no irregular jumps or frame-to-frame jitter. Furthermore, the stability and normalization analysis confirmed that the pose estimation was reliable, indicating that the issue is not related to the input data but rather to the model itself. While it is possible that only a subset of clean samples was selected during the analysis, this is improbable given the random selection process.

The pose estimation approach used in this architecture is frame-based, meaning it does not perform explicit temporal tracking of the human body across frames. As a result, MoveNet Thunder processes each of the 50 frames independently, treating them as individual images rather than as part of a continuous sequence.

This means that the model does not explicitly link body parts across frames over time. Consequently, the keypoint data fed into the model represents a sequence of independent poses rather than a fully continuous motion representation. This limitation may also help explain why the LSTM did not provide significant benefits, as temporal information is likely only partially preserved. Using a pose estimation approach that incorporates temporal tracking could potentially improve the model's ability to learn motion patterns, leading to better generalization and improved accuracy.

Very few individuals perform a squat in exactly the same way. Factors such as timing, speed, starting position, and other biomechanical variations influence how a squat is executed. Consequently, a correctly performed squat can vary between individuals, meaning that correct technique does not necessarily appear identical across different people.

Since the model lacks strong temporal linking and relies more on pose snapshots rather than true motion patterns, this may help explain why performance declined when new participants were introduced. In this context, the same squat does not necessarily correspond to identical joint trajectories across individuals, making it difficult for the model to generalize. This may also explain why the model tends to associate certain participants with front or goblet squats rather than back squats, as their joint trajectories may more closely resemble those patterns.

A potential way to address this limitation, while still using the same pose estimation model, is to increase the size and diversity of the training dataset. Incorporating greater variation in subjects may help the model learn more generalized movement patterns and improve performance on unseen participants.

Another factor to consider is that all participants, including the author, wore similar tight-fitting black clothing during data collection. While this is unlikely to explain the observed generalization failure, it may still influence model performance when applied to more diverse real-world scenarios.

For example, variations in clothing, such as loose garments, patterned fabrics, or multiple colors. Could affect pose estimation quality and, consequently, the extracted keypoints. Investigating the model's performance under such conditions would be a valuable direction for future work.

When considering future work, introducing 3D pose estimation instead of 2D pose estimation could be an interesting direction. Incorporating depth information may provide additional insight into movement patterns and potentially improve model performance. However, it is unclear whether this would lead to significant improvements, given that the current dataset is recorded from a side-view perspective. While using multiple camera angles could enable more accurate 3D pose estimation, this would move away from the original goal of developing a simple side-view squat feedback system that can be used in a mobile application.

It should also be noted that the bar used in this study was a plastic stick intended to simulate a barbell. This setup did not replicate the presence of external load. Introducing a real barbell with weight may influence both movement execution and pose estimation. In a side-view recording, the barbell and weight plates could occlude certain body parts, potentially affecting the visibility and accuracy of keypoint detection. This may introduce additional challenges for the model when applied to more realistic training scenarios.

As of now, the model performs well in controlled settings where the same participants are present in both the training and test data. This makes it suitable for applications such as

personal tracking and same-user feedback. However, the system is not yet ready for real-world deployment. Its performance declines when new participants are introduced and would likely degrade further in more variable real-world conditions.

Overall, these findings suggest that the primary limitation of the system lies in its inability to generalize across different individuals and real-world conditions, rather than in the model architecture itself.

5

Conclusion

5.1 Introduction

This project presents a tool designed to provide automated feedback on squat exercises by analyzing pre-recorded video data. Improper execution of exercises can lead to injury, often due to a lack of knowledge or access to proper guidance. While personal trainers can provide such guidance, they may not always be accessible or affordable.

This research addresses this challenge by exploring the use of pose estimation and deep learning to deliver accessible and user-friendly feedback on squat performance. The proposed system combines pose-based feature extraction with machine learning models to classify squat type, evaluate posture, and generate corrective feedback.

The methodology involved collecting and preprocessing a structured dataset, extracting key-point data using MoveNet Thunder, and training models for classification tasks. The system further integrates a language model to generate natural and actionable feedback based on predefined templates. Validation procedures were applied to ensure consistency and reliability in both data processing and model predictions.

5.2 Summary of Research

The research began with a review of existing work on similar architectures, focusing on how these systems operate and the rationale behind the models used. Based on this review, a

methodology was developed to design the proposed system.

The first step involved creating and collecting a dataset for training. This was achieved by recording side-view squat videos in a controlled environment. The dataset was then augmented using techniques such as zooming, shifting, and scaling, effectively doubling its size and introducing greater variation for training.

Next, the pose estimation model MoveNet was used to extract keypoints from the training data. All recordings were standardized to sequences of 50 frames to ensure consistency across samples.

Two models were then developed: a 1D-CNN model and a 1D-CNN combined with an LSTM layer. This allowed for a direct comparison to evaluate whether the inclusion of temporal modeling improved performance.

Finally, the models were evaluated using precision, recall, and F1-score, along with confusion matrices to assess classification performance.

5.3 Research Objectives

The primary objective of this project was to design and develop a tool capable of providing real-time feedback to individuals performing squat exercises.

This objective was partially achieved. The developed artifact is able to process side-view recordings of squats and generate feedback for users. However, a key limitation is its inability to generalize effectively to new individuals beyond those included in the training data.

Based on the experiments conducted, the system performs reliably for individuals it has been trained on, but its performance decreases significantly when applied to unseen participants.

The secondary objectives were achieved, as demonstrated by the developed artifact and the supporting literature review.

5.4 Research Contribution

This research contributes to the field of pose-based human movement analysis through the development and evaluation of an automated squat feedback system.

The first contribution of this research is the development of a functional artifact capable of analyzing squat recordings. The system integrates pose estimation, deep learning models, and a large language model-based feedback mechanism. Although the system has certain limitations, it demonstrates a working approach that can serve as a foundation for further

development and improvement. Another contribution of this research is the experimental comparison between a 1D-CNN model and a CNN-LSTM model. The results provide insight into how the two architectures perform relative to each other in the context of squat classification.

In addition, this study offers a discussion of potential reasons why the inclusion of the LSTM layer did not lead to improved performance. These findings provide guidance for future research, particularly regarding when temporal modeling may or may not be beneficial in similar applications.

The most significant contribution of this research is the identification of generalization as the primary limitation of the system. The findings indicate that the model architecture itself was not the main cause of poor performance. Instead, the inability to generalize across different individuals was the key challenge. This research further provides a discussion of possible explanations for this limitation and outlines factors that may help future researchers address similar challenges in related projects.

The final contribution of this research is the finding that larger, more distinct movement deviations are easier to classify. Posture classification is an example of that, but more subtle variations, such as squat style are harder to classify. This highlights that posture classification benefits from clearer and more separable features, whereas style classification involves finer distinctions that are more difficult to capture using pose-based representations.

5.5 Future Work

The main limitation of this research is the dataset, which is relatively small and consists only of recordings from the author. Future work should therefore include data from a larger and more diverse group of participants. This would introduce greater variation and improve the model's ability to generalize across individuals.

Additionally, incorporating recordings with variations in clothing, color, and environmental conditions would further enhance the robustness of the model and improve its suitability for real-world applications.

As the dataset was not recorded using a real barbell with weight, this should also be considered in future work. Incorporating a loaded barbell would provide a more realistic representation of squat execution. However, this may introduce additional challenges, particularly in side-view recordings, where the barbell and weight plates could occlude certain body parts and affect keypoint detection. Addressing these challenges would be important for improving real-world applicability.

Another direction for future work is to explore alternative pose estimation approaches. This includes models that provide 3D keypoints, particularly if future research moves beyond side-view recordings, as well as models that incorporate explicit temporal tracking rather than

processing frames independently. Such approaches may improve the consistency of motion representation across frames and enhance the effectiveness of temporal models such as LSTM, potentially leading to improved performance in tasks requiring more detailed motion understanding.

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Sample Appendix

INFORMED CONSENT FORM FOR PARTICIPATION IN RESEARCH**Title of Study:**

Real-time squat feedback using pose estimation and machine learning

Researcher(s):

Sindre Heldal, Noroff university

1. Introduction

You are invited to participate in a research study. Before you decide whether to participate, it is important for you to understand why the research is being conducted and what your participation will involve. Please read the following information carefully.

2. Purpose of the Study

The purpose of the study is to develop a tool that provides automated feedback on squat exercises by analyzing pre-recorded videos.

3. What Participation Involves

If you agree to participate, you will be asked to:

- Perform squat exercises on camera.
 - The expected duration of your participation is approximately [10-20min].
-

4. Voluntary Participation

Your participation in this study is entirely voluntary. You may refuse to participate or withdraw at any time without giving a reason and without any consequences.

5. Risks and Discomforts

This study involves minimal risk. However, possible discomforts may include:

- Training and being on camera.
-

6. Benefits

Figure A.1: Participant 1

While there may be no direct benefit to you, your participation may help contribute to knowledge in the field of Applied data science .

7. Confidentiality

All information collected in this study will be kept confidential. Your identity will not be revealed in any publication or presentation of the results. Data will be stored securely and accessed only by the researcher(s).

8. Data Use and Publication

The data collected may be used in academic publications, presentations, or future research. All data will be anonymized.

9. Contact Information

If you have any questions about this study, please contact:
Sindre Heldal

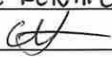
If you have concerns about your rights as a participant, you may contact:
Sindre Heldal

10. Consent Statement

By signing below, you confirm that:

- You have read and understood the information provided above
- You have had the opportunity to ask questions
- You voluntarily agree to participate in this study

Participant Name: ANE RENATE HUOSLEF

Participant Signature: 

Date: 27.04.26

Researcher Name: Sindre Heldal

Figure A.2: Participant 1

Researcher Signature: Sindre Høldal
Date: 27.04.20

Figure A.3: Participant 1

INFORMED CONSENT FORM FOR PARTICIPATION IN RESEARCH**Title of Study:**

Real-time squat feedback using pose estimation and machine learning

Researcher(s):

Sindre Heldal, Noroff university

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-

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5. Risks and Discomforts

This study involves minimal risk. However, possible discomforts may include:

- Training and being on camera.
-

6. Benefits

Figure A.4: Participant 2

While there may be no direct benefit to you, your participation may help contribute to knowledge in the field of Applied data science .

7. Confidentiality

All information collected in this study will be kept confidential. Your identity will not be revealed in any publication or presentation of the results. Data will be stored securely and accessed only by the researcher(s).

8. Data Use and Publication

The data collected may be used in academic publications, presentations, or future research. All data will be anonymized.

9. Contact Information

If you have any questions about this study, please contact:
Sindre Høidal

If you have concerns about your rights as a participant, you may contact:
Sindre Høidal

10. Consent Statement

By signing below, you confirm that:

- You have read and understood the information provided above
- You have had the opportunity to ask questions
- You voluntarily agree to participate in this study

Participant Name: PER VETLE HØIDAL

Participant Signature: 

Date: 27/4 - 26

Researcher Name: Sindre Høidal

Figure A.5: Participant 2

Researcher Signature: Sindre Haldal
Date: 27.04.26

Figure A.6: Participant 2

B

Sample Appendix

The full source code for this project is publicly available on GitHub at <https://github.com/Maklok/REAL-TIME-SQUAT-FEEDBACK-USING-POSE-ESTIMATION-AND-MACHINE-LEARNING>.

Word count metrics

NUC Bachelor Project Word Count:

Total Sum count: 15012 Words in text: 14509 Words in headers: 162 Words outside text (captions, etc.): 341 Number of headers: 74 Number of floats/tables/figures: 40 Number of math inlines: 0 Number of math displayed: 0 (errors:2) NOTE: References are excluded.